Mahindra & Mahindra

QDMS

Functional Design Specification (FDS)

PROJECT #: M&M-REIN SOLUTIONS-QDMS FDS-10-October-18-V1.0

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1. Introduction

This document defines the Quality Data Management System Design Specification (QDMS-DS) for a Drona system designed to control the information flow for the manufacturing of Mahindra AFS Plants. Manufacturing IT is responsible for implementing its Drona system for Mahindra manufacturing plants REIN SOLUTIONS will be working as consulting partner for M&M for implementing its QDMS system. To avoid handling multiple documents, the design specifications may be divided into separate sections representing QDMS requirements.

1.1. Purpose

Quality module is an integral part of Drona ecosystem. This functionality to be tightly integrated with error proofing systems, equipment (conveyor PLC, machines PLC, digital tools, etc.), manufacturing processes. This document provides the basis of the design of the system and is used to verify and validate the system during the testing, ensuring all the required functions are present and that they operate correctly. This document will also serve as the basis for the development of a Quality System Design Specification document for final system acceptance of the system.

1.2. Scope and Key Objectives

The ultimate design of the quality functionality is to develop and deployment of solution for all Manufacturing Shops / Plants. This document only includes design and details of configurable functionality to be developed and deploy across Mahindra AFS.

Manufacturing Quality function plays a vital role to ensure outgoing quality of vehicles. Comprehending & analyzing vehicle build concerns is crucial for effective manufacturing quality function.

QID / QDMS / QP is an acronym for Manufacturing Quality Defect Monitoring & Recording System. It is a system which collects enables data logging, analyses it and gives output to the user in required format (e.g. graphs, pie charts, communication to source etc.). For any system, it is necessary that all data related to whole system should be available on a single click. While users of system avail themselves to the above benefits, they also report that it standardizes the way of logging the defect data in system using user friendly & multi lingual user interfaces.

Drona quality functionality provides real-time analysis of defects captured during manufacturing process to assure proper product quality control and to identify problems requiring attention. It may recommend action to correct the problem, including correlating the defect phenomena, actions & results to determine the cause.

Manufacturing Quality function

- Quick feedback from downstream process to previous process.
- Communication to concern agencies such as SQA, Paint Shop & Body Shop, etc.
- Concern prioritization & analysis.

Manufacturing function

- Monitoring of shop indicators such as Rework Time, STR, FRC, FBO, etc.
- Monitoring of BWT indicators i.e. IO, IC, ID & IQ

Features –

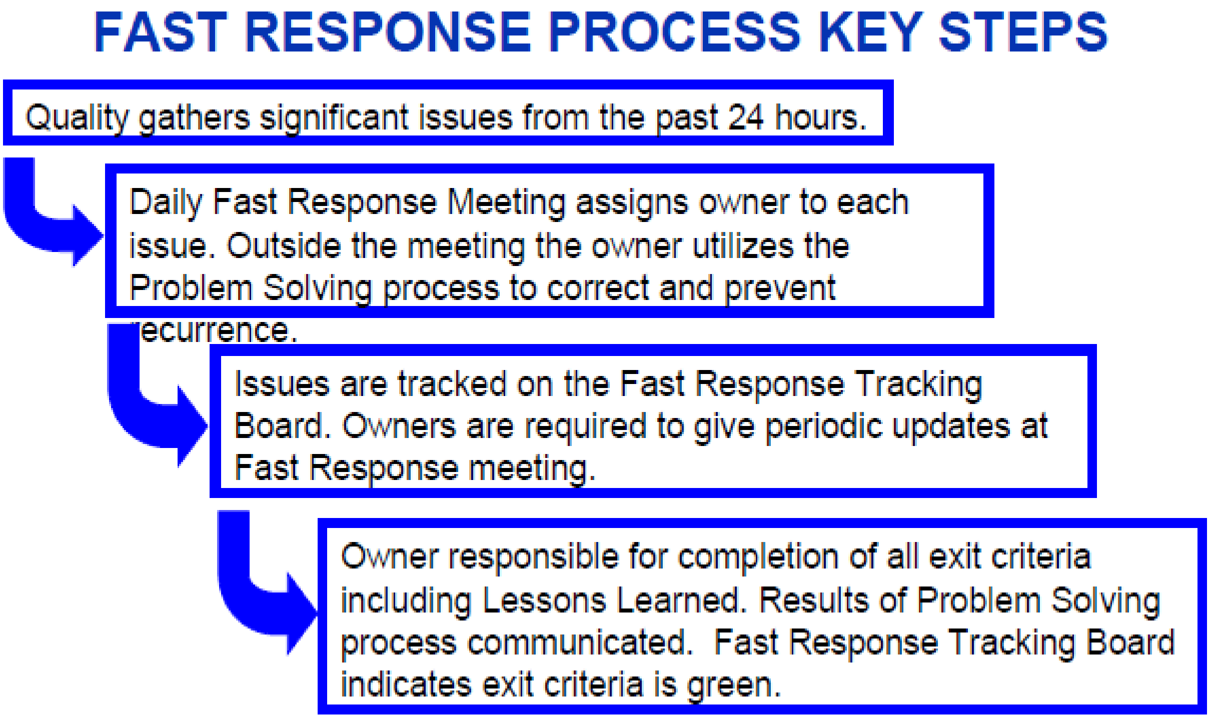
- Convenient data collection
- Real-time data availability
- Online defect communication to source
- Ease of data compilation
- Faster analysis of Top concerns & online reporting

- Paperless manufacturing
- Non-conformance management
- Statistical sampling plans
- Test, Inspection, Defects and Repair
- Defect tracking by location / source
- Parametric data collection
- QDMS is used for Part, Source & Defect wise Data Management
- Online monitoring system for IO, IQ, IC, ID
- Real time feedback to Source BWT / Shop

Ease of data compilation

faster analysis of Top concerns & online reporting

- Quality Indicator wise reports
- Straight through ratio (STR)
- Reporting of high occurrences of defects
- Shop performance reports
- Query based user friendly report generation is possible using various criteria.
- Ease of MS-Excel downloads for making more analytics



1.3. Key agreements

1. New requirements will be reviewed under change management of the Project.
2. The reader of this document is aware of basic automotive manufacturing terms.
3. The reader of this document is also aware of QDMS terminologies.
4. The reader of this document is also aware of QDMS basically.
5. Network will be made available with proper connections and good speed.
6. Required hardware will be made available in timely manner and in required quantity.
7. Environment for development and deployment of Rein solutions will be provided by Mahindra.

2. MES Solution

2.1. Design Agreements

1. Shop Network will be stable and highly available all the time.
2. Shop will be maintaining devices as per the recommendations in them respective user manuals.
3. Failure of one of the systems that integrates to the MES will have an impact on the functioning of the MES. Care has been taken to design work around in case of production Critical requirements.
4. MES and all associated components of the MES will be maintained as per recommendations provided by REIN SOLUTIONS.
5. Operators who interact and use the MES will be trained before they start using the system. REIN SOLUTIONS responsibility will be to train users.

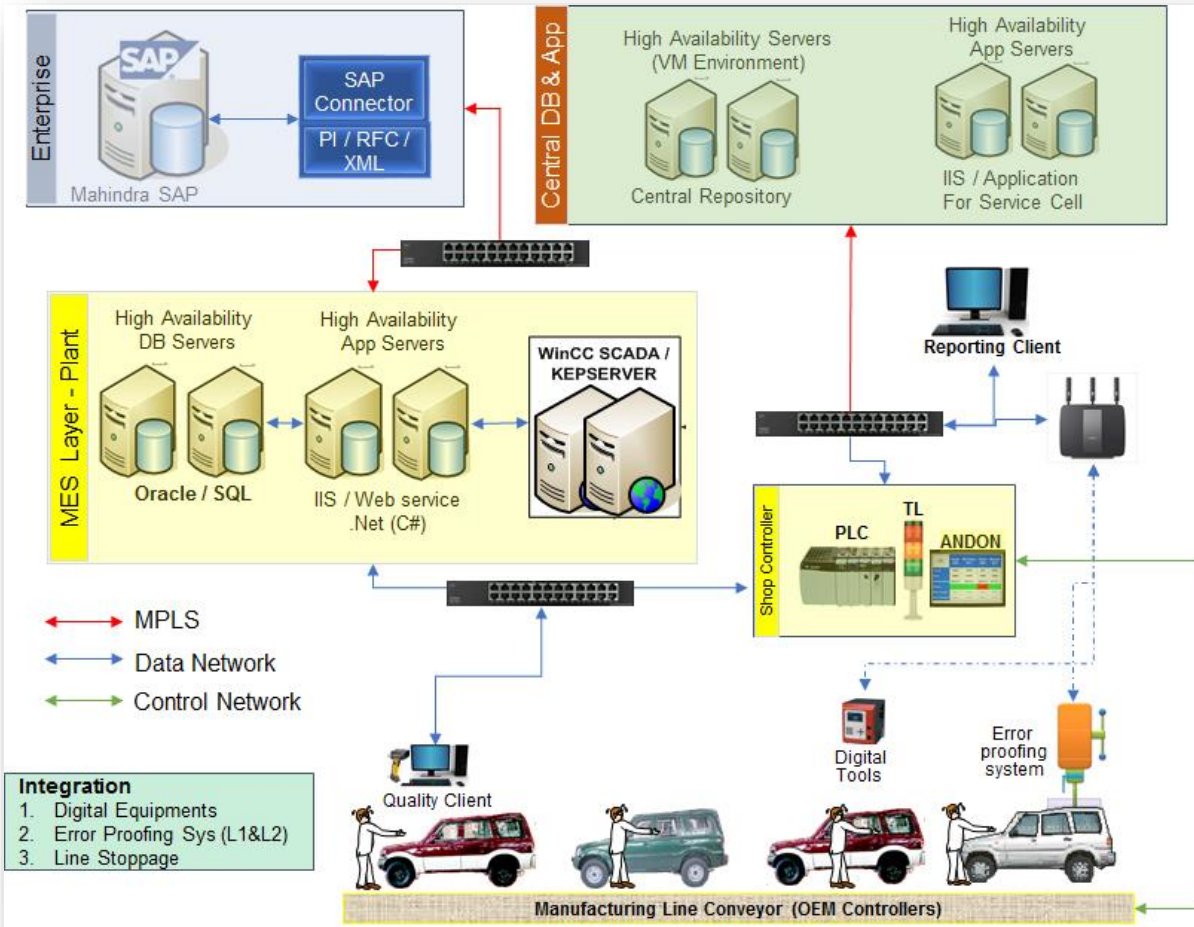
3. M&M, QDMS requirement summary

Below is the matrix of requirement for Chakan Plant:

Module	Functionali	Quantity
QDMS Screen	Operator login and capturing defects against vehicles	**1**
Concern Selection Screen	Operator will be able to select the concern from the repeated concern list.	**1**
Offline Rework Screen	Clearing the defect offline	**1**
Pop-up window	To show the defect which is related to the specific stage	**1**
Defect List	To show the acknowledged defect from pop up window	**1**
Variant Checklist	To show the checklist against model	**1**
Concern Bank Screen	Online Monitoring with actions and closing status	**1**
Auto Mail Form	Form should be provided to send the mail to concerned stakeholders	**1**
Auto Mail Facility	The auto mail should be sent to at the end of the shift	**1**
Reports	Various reports should be provided day wise, month wise, yearly wise	**1**

4. Solution Overview

4.1. System Architecture



NOTE: The database used to develop this functionality will be MS SQL Server.

4.1.1. Server Specifications- Needs to be filled by M&M-IT team

S.No	Server	Server Name	IP Address	OS	Software

4.1.2. PDS (DB)

The **Production database** is an online transaction processing (OLTP) database that records and stores all data collected from Plant Operations. This data is store in database side and we can access data from using SQL server Query.

4.1.3. Reporting Server

Reporting Server provides unified access to all MES data sources, and produces web-based reports, such as dashboards, trends, X-Y plots, Excel reports and more that can be used by manufacturing operators, engineers, supervisors, management and executives throughout a plant — to manage cost, quality, production, assets and resources more effectively.

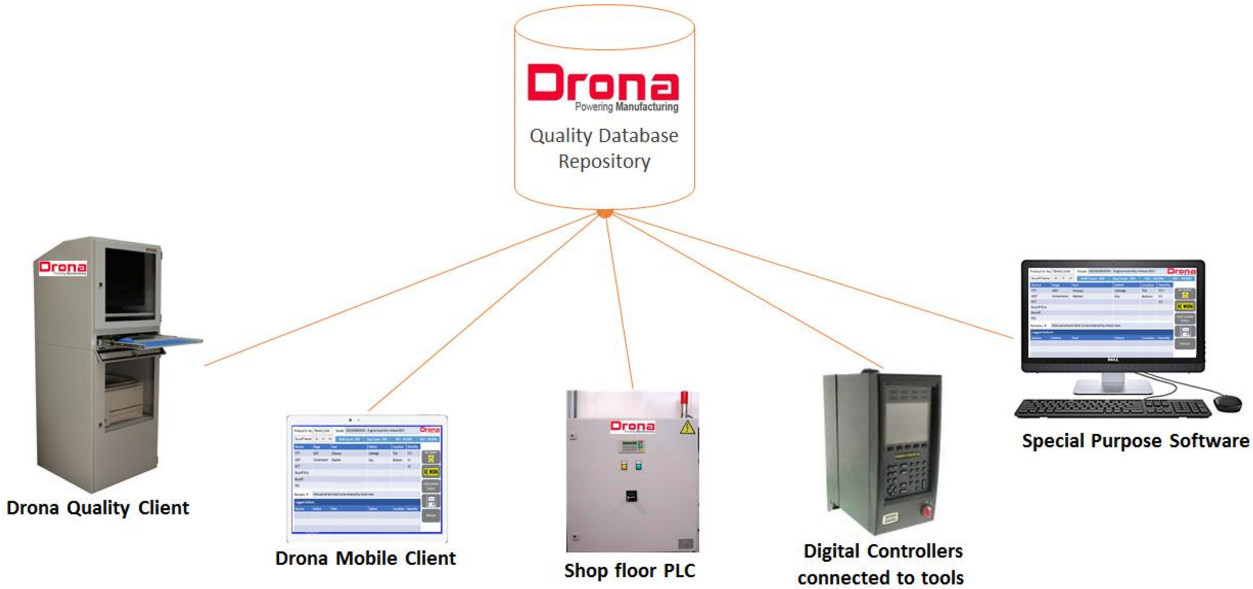
4.1.4. Application Server

The Plant Operations **application server** is the functional representation of the transactional system that feeds data to the Production database. Users connect to application servers to build and run the production floor applications, and third-party applications connect to the application servers to load data to the Production database.

4.2. Server Sizing- To be filled by M&M, IT

Server Name	Qty	Description
Application Servers		
DB Server		
Application services		
Reporting Server		

4.3. Source of Manufacturing Quality Data

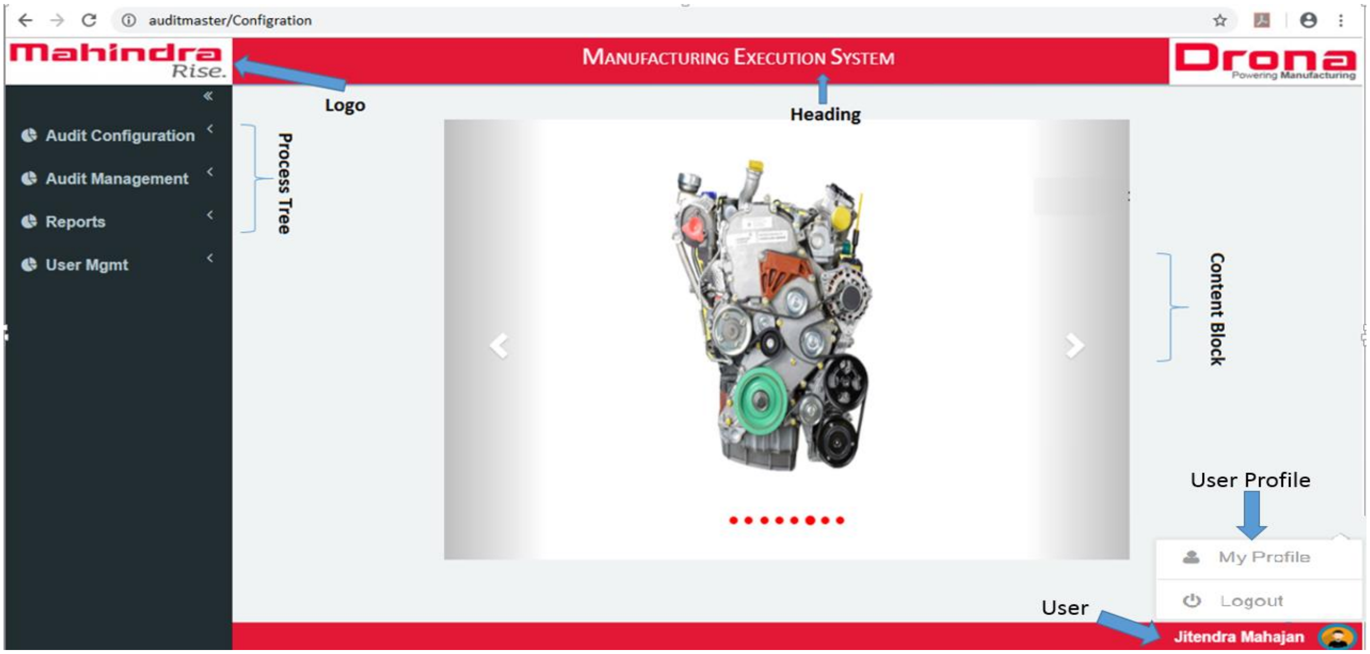


4.4. Products Used

Product	Version
Angular	4.0
MVC 5 (Razor)	5.0
Google Chrome	Higher than version 56.0
SQL SERVER	SQL 2012 and Above
SSRS for reporting	SQL 2012
Web Services	
IIS server	Above than windows 7

4.5. User Interface Standards

Below screen is standard which will be followed for user interfaces:



4.5.1. Assumptions

The following assumptions hold true:

- Mahindra logo image is displayed on the top left corner of the application title window.
- The client screen should support resolution of 1024x768. This can be changed based on the screen choose in shop floor.
- The client should have Chrome 55 or higher versions. The application should run smoothly on other browsers like Firefox, Safari, and Chrome etc.

4.5.2. Layout

Title Area – Area that contains the main information about the application. The Title Area will have the following information displayed always.

- Mahindra logo – Mahindra logo image, centered top-to-bottom on the far-left side.
- Application name – Manufacturing Execution System, centered top-to-bottom, just to the right of the Client logo image.
- User - User Name of the person who is logged into the application, located just below the home logo image which located below the client logo image and left-justified with the site id.
- Main Menu - Presents the application's top level of options.
- Screen Header – The main menu and submenu (if applicable) in which the selected page is located.
- Page Title – The title of the current page being displayed to the user.
- Body – Area that contains the functionality being invoked.

4.5.3. Font and Style

Below is the list of specification for the style that are applicable for the controls and the content that appears in the MES Application.

- Type Face – The font should be same on pages as per the reference theme. Ex. Calibri(Body) as the standard font for English.
- Type Size – Font size should be consistent. The size of the font/font height will be 12px.
- Case – All text should consist of upper and lower case. The first letter in each word that comprises a label should be capitalized. Acronyms will be displayed in all capitalized letters
- Emphasis - Any emphasis that needs to be applied to text will be performed through varying the font's weight and size.
- Alignment – Titles, headings, body text and input or display fields will appear left justified in relation to themselves. Labels will appear right-justified. Margins and or outline boxes may be used to delineate logical groupings of related information.
- Screen Background Color: Screen should be Windows Grey RGB (212,208,200). This could be changed if
- M &M has specific color pattern.

4.5.4. Screen Header

The screen header will include a unique name describing the current window, screen or entry form. For example, if user initiated "Manage Users" menu item from the main menu. "Administration," the screen header will only display "Manage Users" as the title.

4.5.5. Home Page

The "Home" page displays all the main menu items and their associated submenu items to the lowest level of expansion with links. This page provides an optional navigation point for the user to launch screens directly as opposed to launching screens from the main menu area. Depending on user access security, only the appropriate main menu items and their associated submenu items are presented to user.

4.5.6. Popup Windows

Pop-up windows provide the ability for the user to process an action while keeping the associated information available for review. After the user processes the pop-up window, the user will be returned to the application. Should a function within the application require more, approval must be obtained. Any verification questions will use a standard pop-up window with Yes and No buttons.

4.5.7. Text Boxes

Text boxes may be editable or non-editable. Editable text boxes enable the user to input free-form text into the field. These editable text boxes are displayed as an outlined rectangular box, inside which the user can enter text. Non-editable text boxes will serve the purpose of displaying text. The non-editable text box will not display the outlined box. It restricts the user from typing text into it.
The maximum size of a text box will be limited to the size of the corresponding database field. A single-line text box will be used if only one line of text is required. A multi-line text box may be used if more than one line of input is required.

4.5.8. Push Buttons

A push button is a small, rectangular object used to perform an action. For example, the Save and Clear buttons on a dialog box are push buttons. The push buttons must be unique on each screen and must be the same size on the same screen regardless of their label's size. The enter key must not activate any buttons, unless that button has been highlighted on the screen. The Save button on clicking will perform a successful operation whereas the Clear button will clear all the user inputs and empties the control associated in that form. Push button can have image on it to indicate purpose if required. That image needs to remain consistent across the application.

4.5.9. Drop down List

- Drop-down List boxes display a non-editable list from which a user can select an item. The drop-down list box should be used over radio buttons when two or more items need to be grouped or when the space is tight.
- Drop-down list where default data is required should be specified as default item. For user, driven selection Drop down List should always select “Select” text as default selection. For drop down list where user can choose to keep selection empty, option of “No Selection” should be available.

4.5.10. List Boxes

Like drop-down lists, a list box contains a static, multiple line list from which a user can select an item. The list box will be used when the list of possible items could grow longer than twenty items. A scroll bar will be located on the right-hand side to search through the list. A multi-select list should be used when selecting multiple items in the list.

4.5.11. Radio Buttons

Radio buttons are used to select one mandatory option amongst several options. The radio button contains a small circle with text on the right-side. When selected, the circle is shaded. Selecting one button in a set deselects the previously selected button, so only one of the options in the set is selected at a time. Radio buttons should be used with discretion and are best suited for representing discrete choice amongst 2 to 3 items or when a choice needs to be qualified further. Radio button should not be used for selection more than.

4.5.12. Standard Buttons

- Distinct buttons are used to differentiate creating records from updating existing records. The Standard Button set consists of Create, Update, Delete, Save, Clear, Cancel, Yes and No buttons. The buttons will be grouped together according to the function being requested. Typically, the Create, Update and Delete buttons appear together on the Manage screens (a list of items displayed in table form).
- Standard positioning of the buttons will create a consistent feel across the application. The group of buttons will be centered within the form. The Standard Buttons will be consistent with the push button characteristics.

4.5.13. Tables

Two different types of tables are used in the application. Simple Tables are used for display purposes only. Complex Tables allow a user to select a row and then click a button to perform an operation. Selected row should be highlighted with a different shade. Functional Primary key for the data in selected row should be the first column.

4.5.14. Simple Table

The simple table will be used when only a few rows of similar information need to be displayed within the system.

4.5.15. Group Boxes

The group boxes in Angular are used for grouping a collection of objects. The title of the group box tells the intention of grouping to the user. Mostly group boxes are used to group radio buttons, which allows the user to select an option among a group of available options.

4.6 Functionality requirement overview:

Audit Functionality	Explanation

QDMS Operator Screen	• Associate will login on the screen.
	• On scanning of barcode on the traveler card, the predefined QID Stage location will get selected automatically.
	• The system will provide the facility to capture defects against vehicle by image based or part based.
	    • Image Based: - User select the Grid on image. Grid will show all parts configured for that grid. User will select the part. The defect configured against part will listed, user can select the defect from the defect list. The grid will have 6 X 6 fixed size.
	    • Part Based: -
	    User select Source->Stage->Part->Defect->Location->Severity (this sequence should be before final buyoff) to enter the defect.
	    User select Part->Defect->Location->Severity->Source->Stage (this sequence should be at buyoff or after buyoff)
	        o Source, stage should have intelligent search, severity if not defined then give drop down to select severity.
	        o The part base, image based would be configured against platform, shop and QID Stage.
	        o Predefined severity and attribution should be shown, if not, give provision to select.
	        o There should be a small button near check box in ''select for rework "column, clicking on it will clear the concern (with system time, reworked by as logged on user)

Concern Selection Screen	• The concern list will be empty at the start of the day shift • There will be a button on operator screen, on clicking of it, the screen will be opened. • Once operator starts inserting the defect, then it will start getting inserted in to the list. • Operator can select the defect from the concern list without going through all the process.
Offline Rework Screen	• User needs to login with token number to open the QDMS form • User will scan vehicle number in text box and against vehicle number, data will be shown on the QDMS screen. • After clicking on the rework button, the data against the vehicle number scanned on QDMS screen will be populated on rework screen. • Relevant data should be filled to close the defect.

Pop Up Window	• After pressing the save button for the vehicle, popup should be displayed if there is any defect at this station. • Operator needs to acknowledge it then the next vehicle can be scanned. • While working on vehicle, if there are any defects get assigned to this station, the pop up won't appear until the save button is pressed. • Interplant facility needs to be provided.
Defect List	• Button should be provided in the footer to display the popup defect list for the day.
Variant Checklist	• Button "Checklist" should be provided on QDMS screen to show the checklist against model. • On click of "checklist" button, the checklist against model should be displayed. • On this form, submit button should be provided. • On clicking of it, the data will be stored against Plant, Shop, Line, Serial Number, Checked, By Whom
Concern Bank Screen	• 2 different tabs should be provided on the form for DWM concern and QCRT concern. • Show top 5/10/15 concerns on the form depending upon the configuration • User will provide root cause, attribution, action, responsibility, target date and status for the specific defect. • Forum dropdown should be provided with 2 options DWM and QCRT • Hyperlink should be provided on 'no of occurrences' column to show the number of occurrences day wise and vin no wise
Auto Mail Form	• System will allow to put mail as per attribute to concern stake holder team. • Draft mail will appear with given mail ID's but person will manually click 'Send' button just to add if he wants anybody to add in mail information or action.
Auto Mail Facility	• At the end of the shift, one auto mail with top 5 concerns = occurrence * Severity should be sent • second mail with occurrence wise should be sent
Report	Following reports should be developed **BIW** • FRC /FBO – ShiftWise, day wise, Month wise, year wise, BWT Wise • Top 5 concerns – Occurrence wise + Severity*Occurrence wise – daily ShiftWise • TCF & Paint Shop RPT related to BIW – Daily, Shift wise, Monthly , Yearly • RFI DPT - (Editable to CQA) • Concentration diagram • Attribution wise concern status • Rework Manhours • Report for number of hold vehicles in the respective areas

	Paint Shop • Top 5 concerns – Occurrence wise + Severity*Occurrence wise – daily ShiftWise • TCF RPT related to Paint Shop – Daily, Shift wise, Monthly, Yearly • RFI DPT • Concentration diagram • Attribution wise concern status • Rework Manhours • Report for number of hold vehicles in the respective areas • Hold Bodies status model wise, date wise, ShiftWise monitoring **TCF** • FRC /FBO – ShiftWise, day wise, Month wise, year wise, BWT Wise, Line wise, model wise • BIQ Pillar wise report • Top 5 concerns – Occurrence wise + Severity*Occurrence wise – daily ShiftWise • RFI DPT • Total RPT (up to WRAP) • Offline RPT • Offline RPT (Functional/Nonfunctional) • Shower FRC • Concentration diagram (Shower + Paint booth) • Attribution wise concern status • Rework Manhours • Report for number of hold vehicles in the respective areas **RFI** • Weighted DPV as per model, Within model- department wise, within attribution wise and model wise • Same above only for V1&V1+

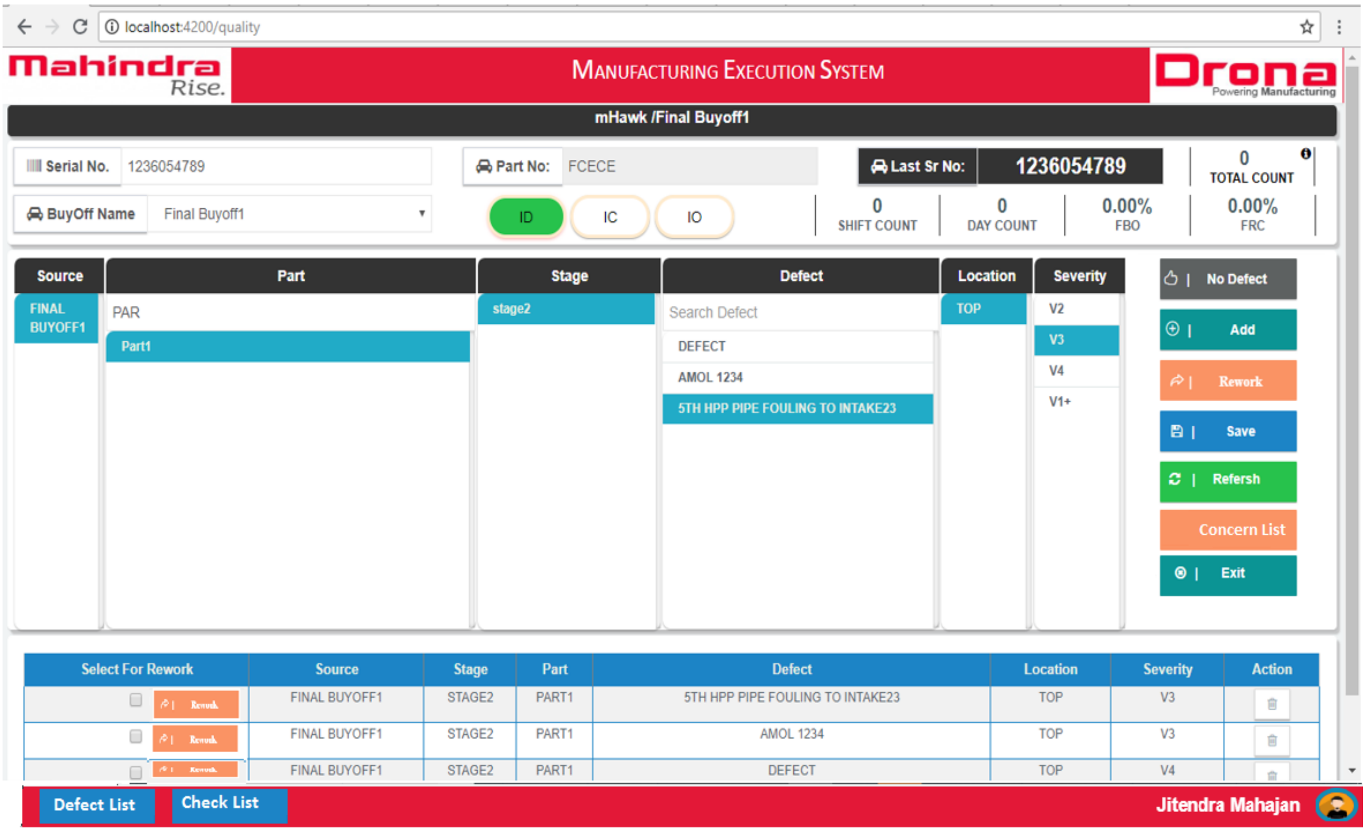
4.7 QDMS Operator Screen

Following screens shown in this section needs to get developed in for QDMS project. There are 2 types of screens (part based and image based) for defect capturing depending on the plant and shop requirements

Sr. No	Plant	Image Base	Part base
1	Nasik, Zaheerabad-AD, Haridwar	BIW, Paint Shop	TCF
2	Chakan	BIW, Paint Shop, TCF	-
3	Igatpuri/MHEL	TBD	TBD

Part Based Screen

Step 1: User will login with the token number. The QDMS (part based or image based) screen will be opened depending on the configuration.



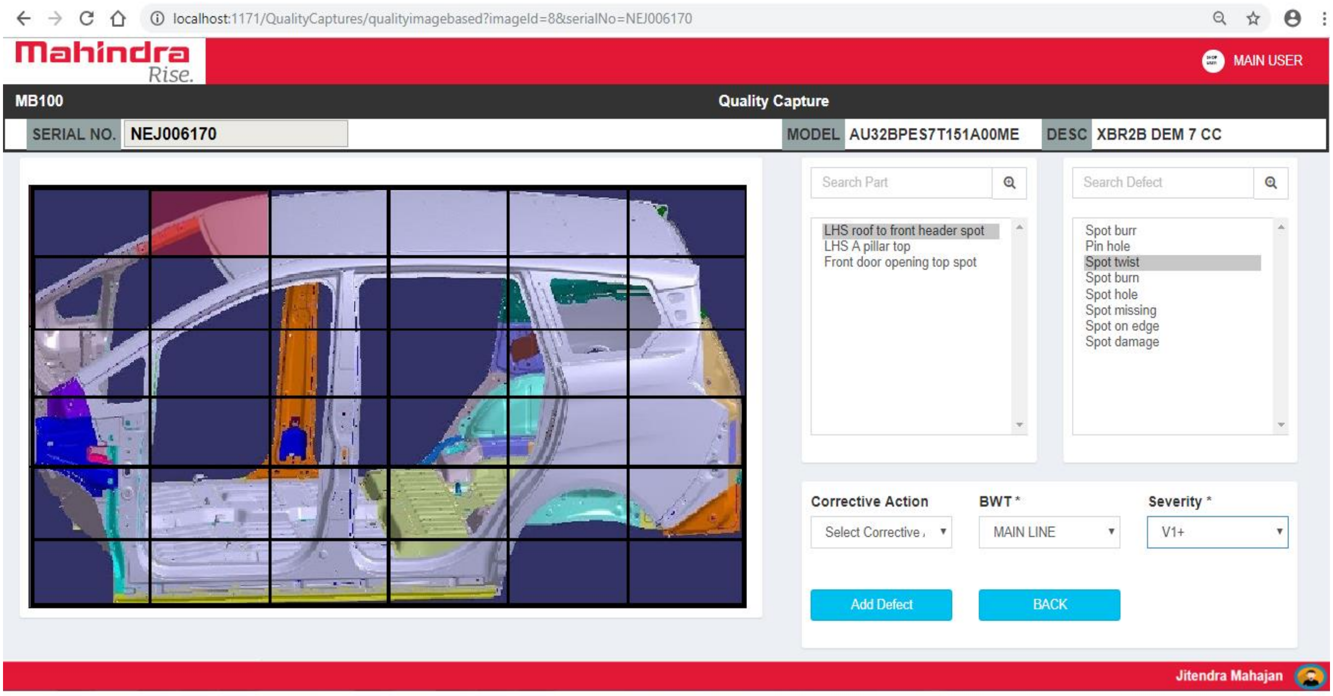
Step 2:

- User will scan vehicle serial number in text box and against vehicle serial number, other attributes and details will be displayed on QDMS Screen.
- User will select source, stage, part, defect, LOC, severity and BWT from drop down. Then user will click on the ADD/Update button then defect will be added in data grid view and user will click on the save button then data will be saved in database.
- There is a 'rework button' available in the 'select for rework' column near check box to clear the defect online. The defect will be cleared after clicking on this button with the system time and reworked by as logon user.

- There is a ‘concern directory’ button available to select the defect from the list which will opened after clicking on the ‘concern directory’ button.
- There is a ‘defect list’ button available on the screen in the footer to see what all defects have been acknowledged on this station

Image Based Screen

Step 1: User will login with the token number. The QDMS (part based or image based) Screen will be opened depending on the configuration.



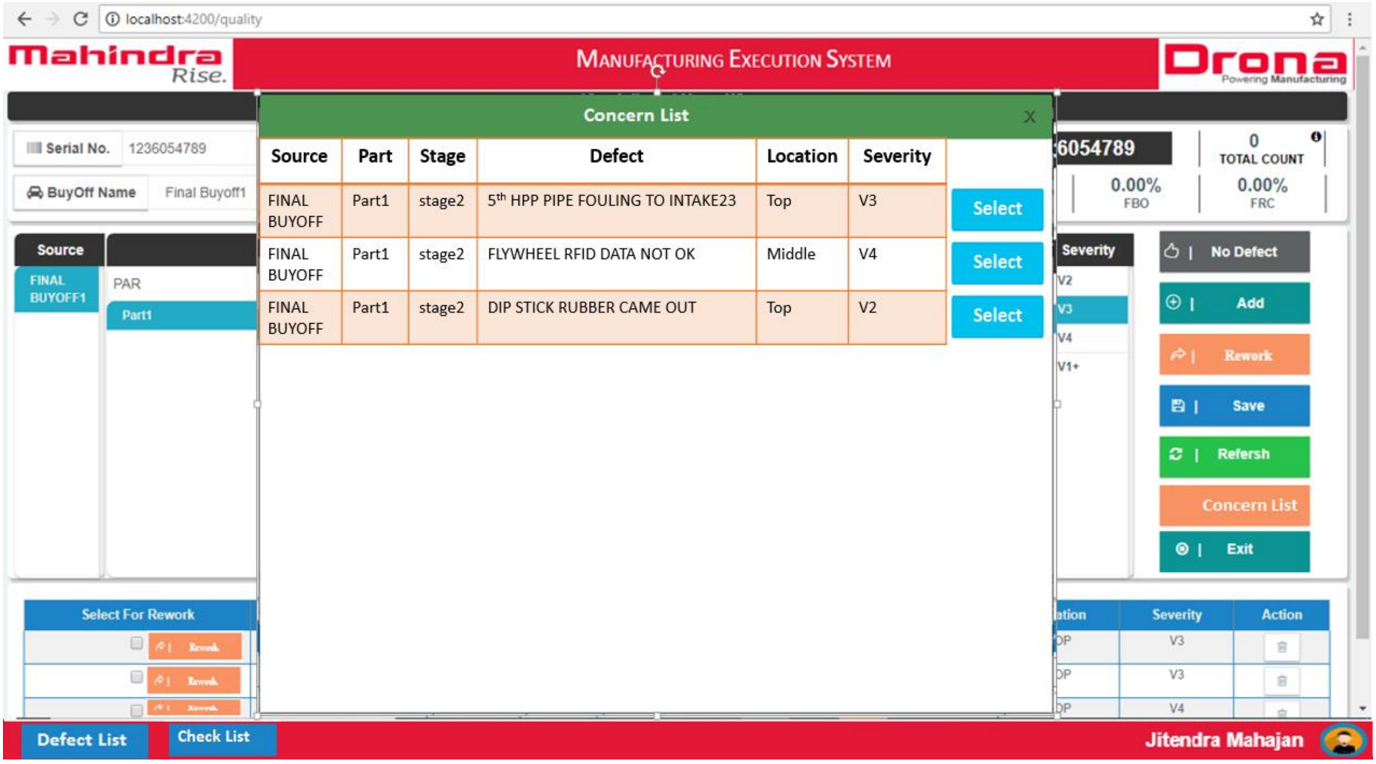
Step 2:

- User will scan vehicle serial number in text box and against vehicle serial number, other attributes and details will be displayed on QDMS Screen.
- User will have the facility to select the image for different views
- Image will be displayed with fix grid size 6X6
- Operator will select the cell from an image
- After selecting the cell, parts configured against that cell will be displayed
- After selecting the part, the defect captured against the part will be shown
- Operator will select the defect and save

4.8 Concern Selection Screen

This section describes the functionality of concern list. Following screen shows how the concerns will be displayed

Step 1: After clicking on the concern list button, the concern list will be shown.



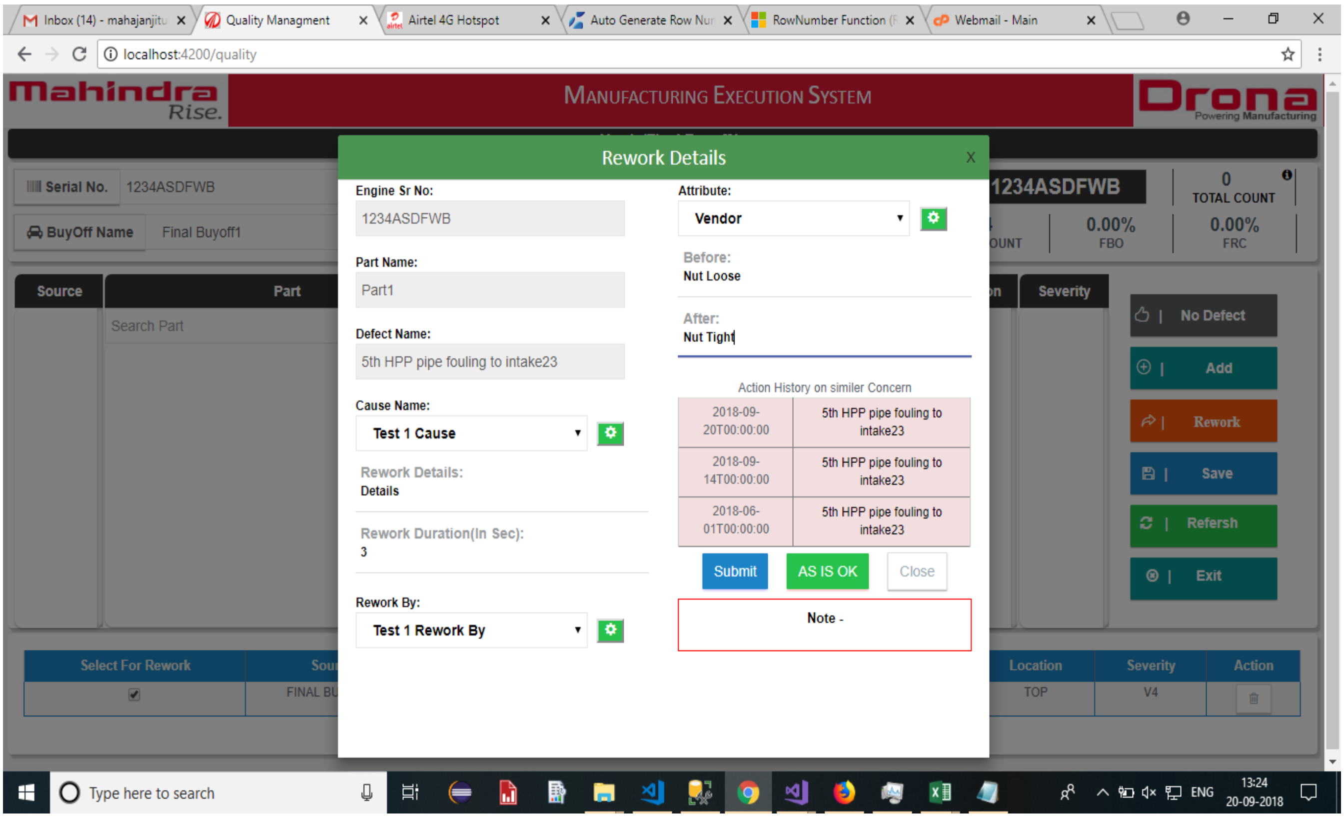
Step 2:

- The concern list will be empty at the start of the day shift.
- After operator starts entering defect against the vehicle, the data will be stored in backend.
- Part and defect will be populated in the concern list screen
- Operator can select the defect from the concern list quickly to avoid all process.
- Operator will click save button on the QDMS screen to save the defect against the vehicle.

4.9 Offline Rework Screen

This section describes how the offline rework will work

Step 1: User will click on the rework button to clear the defects.



Step 2:

- Vehicle number scanned on QDMS screen will be taken to display defect against the vehicle number on rework screen
- Operator will fill the necessary details on rework screen to clear the defect
- After clicking on submit or as IS OK, the defects will be cleared.
- There should be a facility to upload image/video on the rework screen.

4.10 WRAP System Interlock

This section describes how the WRAP Interlock System will work.

Step 1: User will login with the token number. The QDMS (WRAP System) Screen will be opened Depending on the configuration.

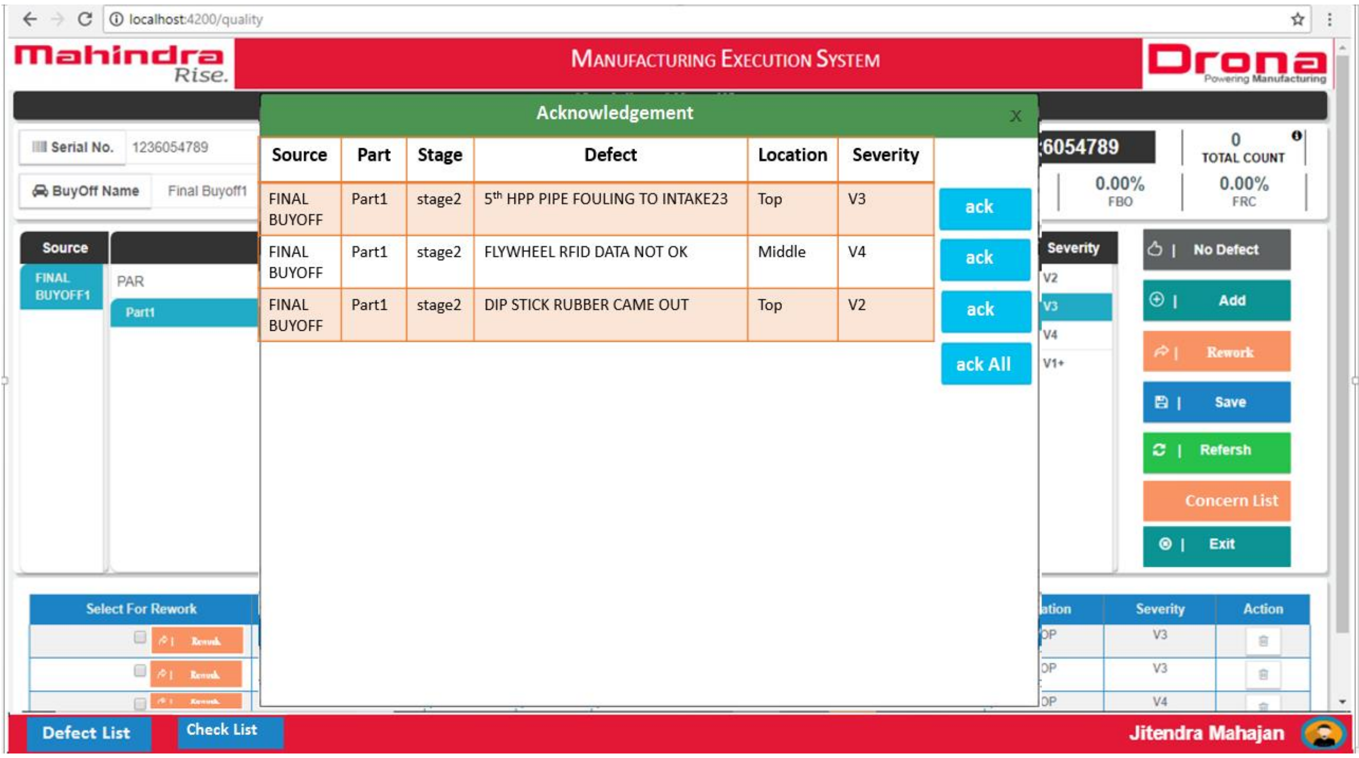
Step 2:

- User will scan vehicle serial number in text box and against vehicle serial number, other attributes and details will be displayed on QDMS Screen.
- Operator will fill the necessary details on WRAP screen to generate the barcode.
- System will not allow to make WRAP 'OK' unless and until all reported concerns in QDMS system are OK.
- WRAP Person can go through the complete rework history against vehicle during final check to do WRAP OK
- There should be a facility to generate WRAP 'OK' barcode on the WRAP screen.

4.11 Pop up Window

This functionality displays the defects generated at this station but not captured.

Step 1: Pop up window will be displayed automatically when there is no vehicle scanned. This depends upon The part and defect configuration.



Step 2:

- After the vehicle has been processed, the popup will be shown with defects.
- The defects displayed at this stage are generated at this stage but not captured.
- Screen with individual 'ack' button for the defect will be displayed.
- Screen with individual 'ack all' button for all defects will be displayed.
- Operator can acknowledge individual defect or all defects at once by clicking the respective button.

4.12 Real Time Quality Communication

This section describes how the Real Time Quality Communication System will work.

Description: - We will configure either host name or IP address to display the real time quality Communication.

- After opening Real Time Quality Communication system all the configured attributes like Line, BWT , Stages, Date ,Shift and all other details will be displayed on QDMS Screen.
- On the first part of the screen we will have following column: concern name, Severity, time, source, occurrence, vehicle serial number, and Acknowledge to display todays concern If the associate acknowledge the defect on popup screen (**4.11**) then the button color on acknowledge column will changed to green if he does not acknowledge the defect then the color will not change.
- On other side of the screen we will display Top five repeated defect/concern of the day in the column: concern name, and the severity of the defect in the severity column and occurrence of the defect how many times in a day the defect has been generated in the occurrence column.

Below screen will be shown

Line	Trim Line	**BWT**	01	**Stages**	01-06
Date	5/7/2018	**Shift**	A		

Today's Concern

Sr.No	Concern Name	Severity	Time	Source	Occurrence	Veh. Sr.No	Acknowledgement
1	Frt. Door & Fender Fouling	V1	11:00AM	RFI	2	XXX	
2	Floor Console Loose	V1	1:20PM	FAI	5	XXX	
3							
4							
5							

Repeatative Concern's

Concern Name	Severity	Occurrence
Tail Lamp to Side Outer Gap	V1	10
3rd Row Grab Handle Loose	V1	8

EN Communication

Tail Door Waist Rail Change

Automatic Steering gear commonisation with Manual Steering gear

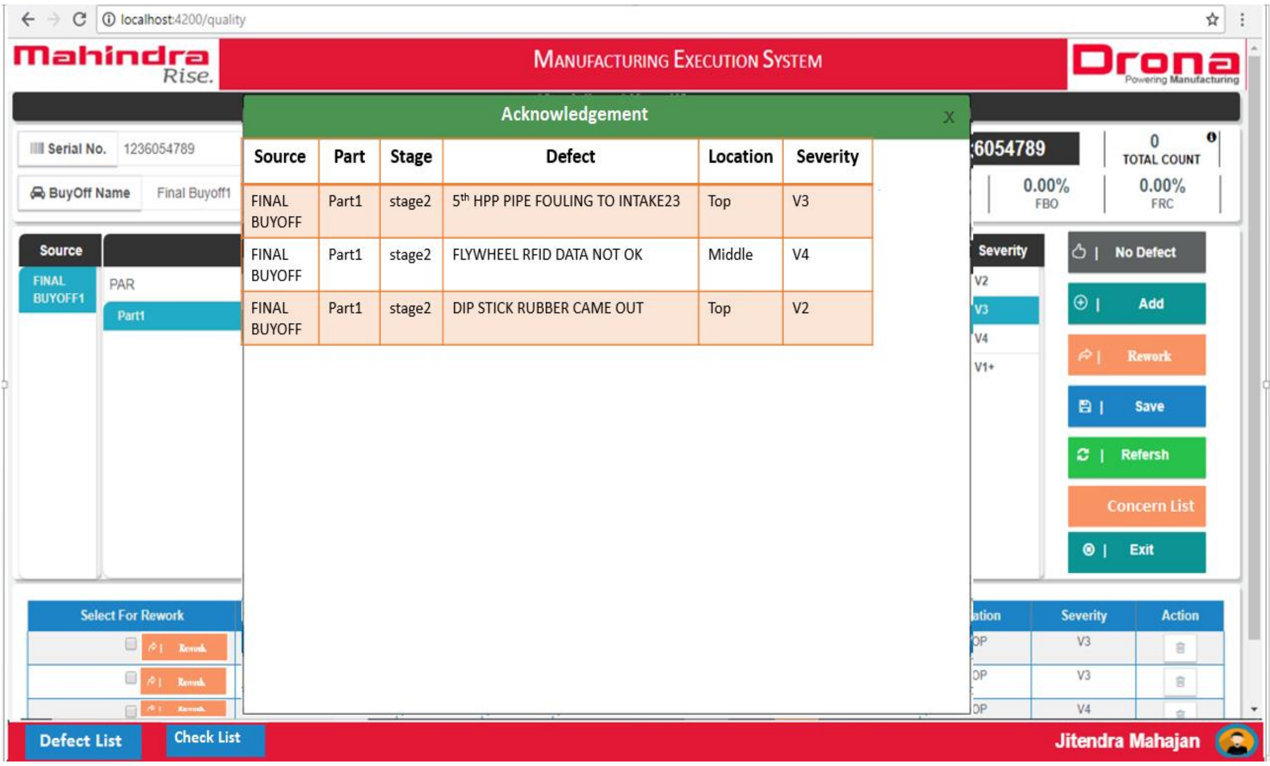
MIS

FRC	75%
FBO	85%

4.13 Defect List

This functionality displays the acknowledged defects.

Step 1: Click on the button provided in the footer.

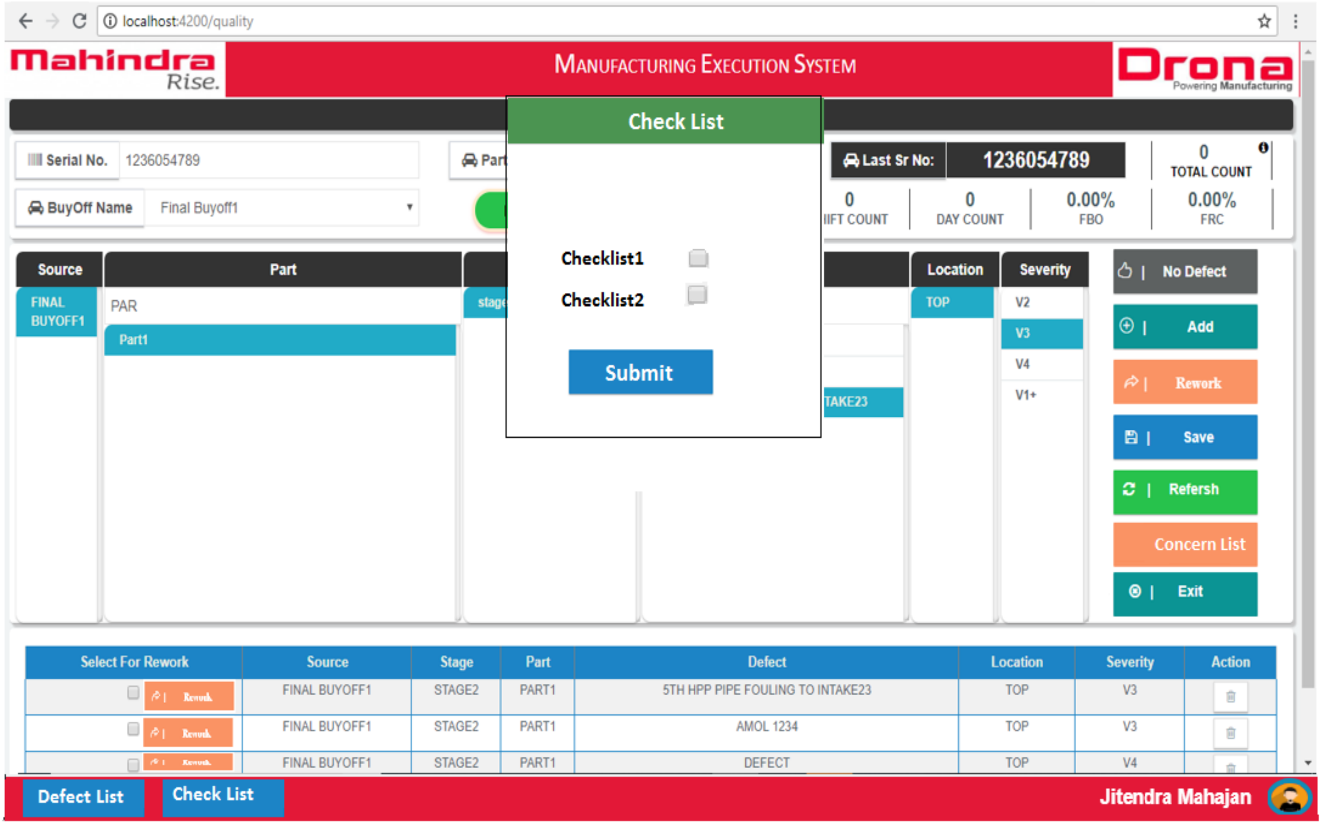


Step 2: Form with defects acknowledged on this stage will be displayed

4.14 Variant Checklist

This functionality displays the checklist against the variant

Step 1: Click on the button will display the checklist against the model.



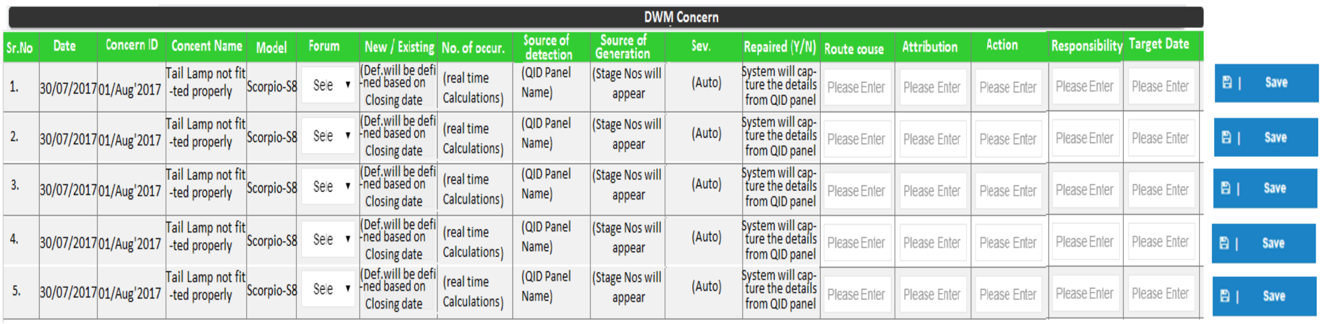
Step 2:

The checklist against the model should be verified.
Operator will click on submit button after verifying the checklist.
Data will be stored for the checklist.

4.15 Concern Bank

This functionality provides the facility for online monitoring and closing status.

Step 1: Click on the separate link to open the online monitoring and closing status form



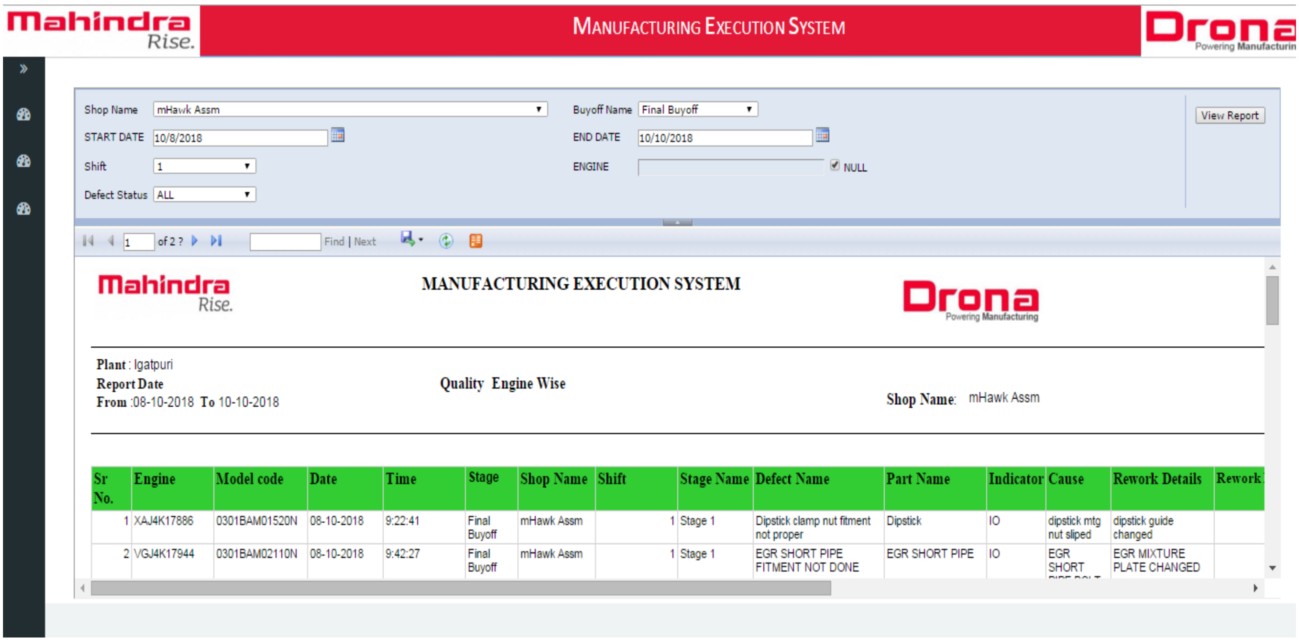
DWM Concern

Sr.No	Date	Concern ID	Concent Name	Model	Forum	New / Existing	No. of occur.	Source of detection	Source of Generation	Sev.	Repaired (Y/N)	Route couse	Attribution	Action	Responsibility	Target Date	
1.	30/07/2017	01/Aug'2017	Tail Lamp not fit -ted properly	Scorpio-S8	Sele	(Def.will be defi -ned based on Closing date	(real time Calculations)	(QID Panel Name)	(Stage Nos will appear	(Auto)	System will cap- ture the details from QID panel	Please Enter	Please Enter	Please Enter	Please Enter	Please Enter	Save
2.	30/07/2017	01/Aug'2017	Tail Lamp not fit -ted properly	Scorpio-S8	Sele	(Def.will be defi -ned based on Closing date	(real time Calculations)	(QID Panel Name)	(Stage Nos will appear	(Auto)	System will cap- ture the details from QID panel	Please Enter	Please Enter	Please Enter	Please Enter	Please Enter	Save
3.	30/07/2017	01/Aug'2017	Tail Lamp not fit -ted properly	Scorpio-S8	Sele	(Def.will be defi -ned based on Closing date	(real time Calculations)	(QID Panel Name)	(Stage Nos will appear	(Auto)	System will cap- ture the details from QID panel	Please Enter	Please Enter	Please Enter	Please Enter	Please Enter	Save
4.	30/07/2017	01/Aug'2017	Tail Lamp not fit -ted properly	Scorpio-S8	Sele	(Def.will be defi -ned based on Closing date	(real time Calculations)	(QID Panel Name)	(Stage Nos will appear	(Auto)	System will cap- ture the details from QID panel	Please Enter	Please Enter	Please Enter	Please Enter	Please Enter	Save
5.	30/07/2017	01/Aug'2017	Tail Lamp not fit -ted properly	Scorpio-S8	Sele	(Def.will be defi -ned based on Closing date	(real time Calculations)	(QID Panel Name)	(Stage Nos will appear	(Auto)	System will cap- ture the details from QID panel	Please Enter	Please Enter	Please Enter	Please Enter	Please Enter	Save

Step 2:

- Operator will be provided with 2 different tabs
    - DWM Concern
    - QCRT Concern
- Screen will display top 5/10/15 concerns on the form depending upon the configuration
- User will fill following fields to take an action on the defect
    - Root cause
    - Attribution
    - Action
    - Responsibility
    - Target date
    - Status
    - Forum
- The button to send email will be provided on the screen, on click of it the Auto mail form will be opened which is described in the next section.
- Hyperlink should be provided on 'no of occurrences' column to show the number of occurrences day wise and vin no wise.

  Following is the sample screen to show the no. of occurrences day wise and vin wise

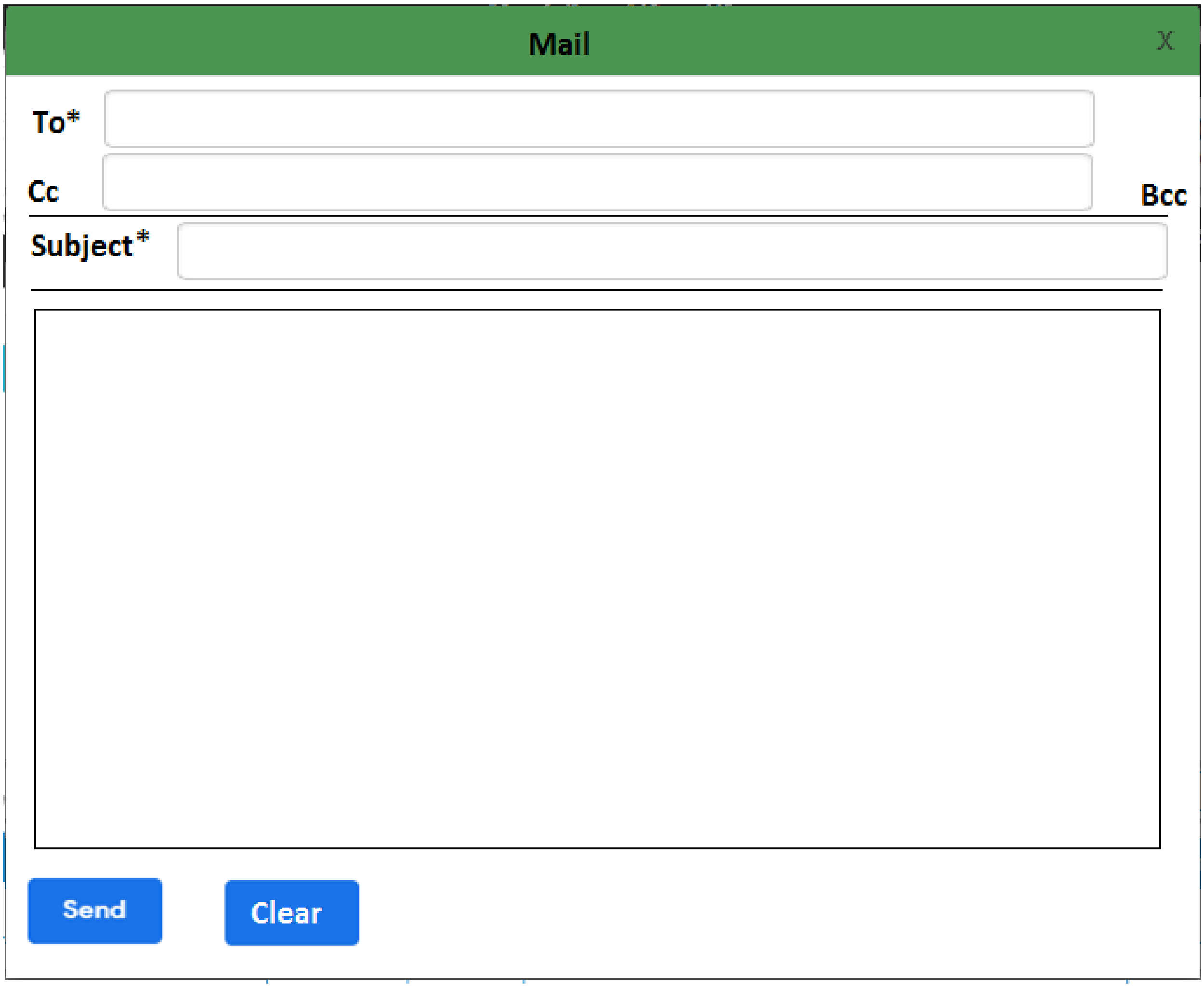


4.16 Auto Mail Form

This form will send the report to concern stakeholders as per attribute in an email.

Step 1: Following form will be opened with To, CC and body

To: Attribute wise stakeholder email addresses.
CC: Facility to add more email addresses.
Subject: To add the subject
Body: The email body with reports contents inside



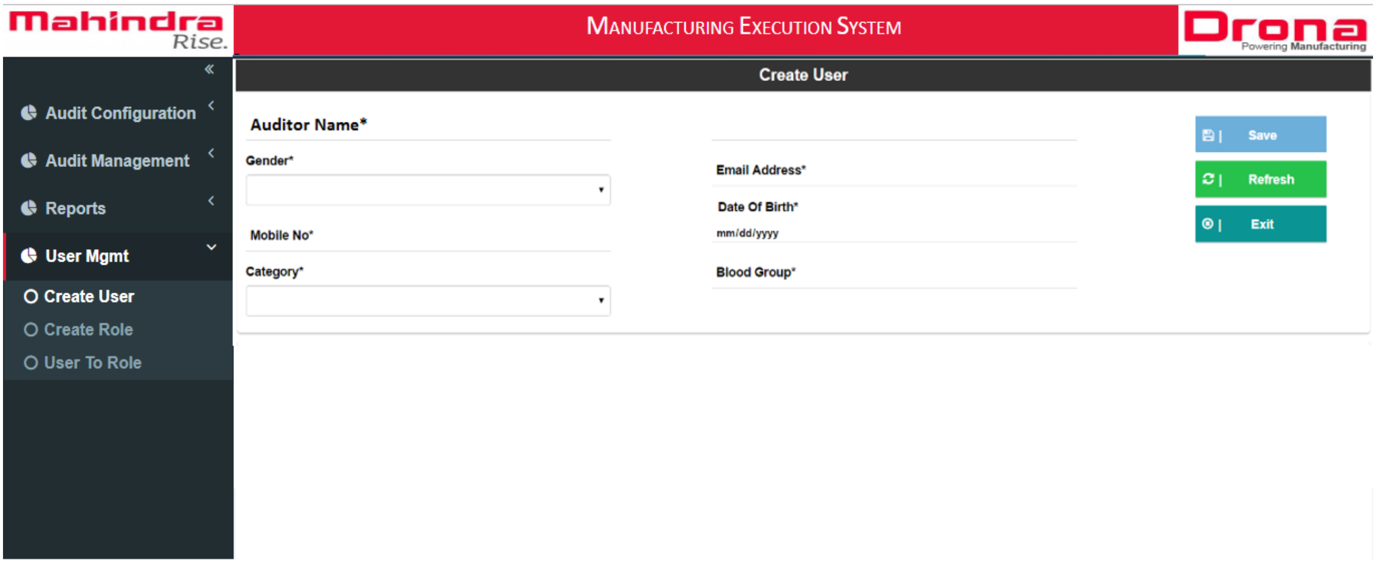
Step 2: On clicking on the send button, the email will be sent

4.17 User Management Configuration

Users are very important person to doing task. When vehicle randomly get selected from yard, auditor login is most important to fill the write up sheet.
After submitting the write up sheet for the audited vehicle, in the show and tell meeting, different users must do different action according to their permissions.

4.17.1. Create User Configuration

This configuration is used to create the master list of users with their details



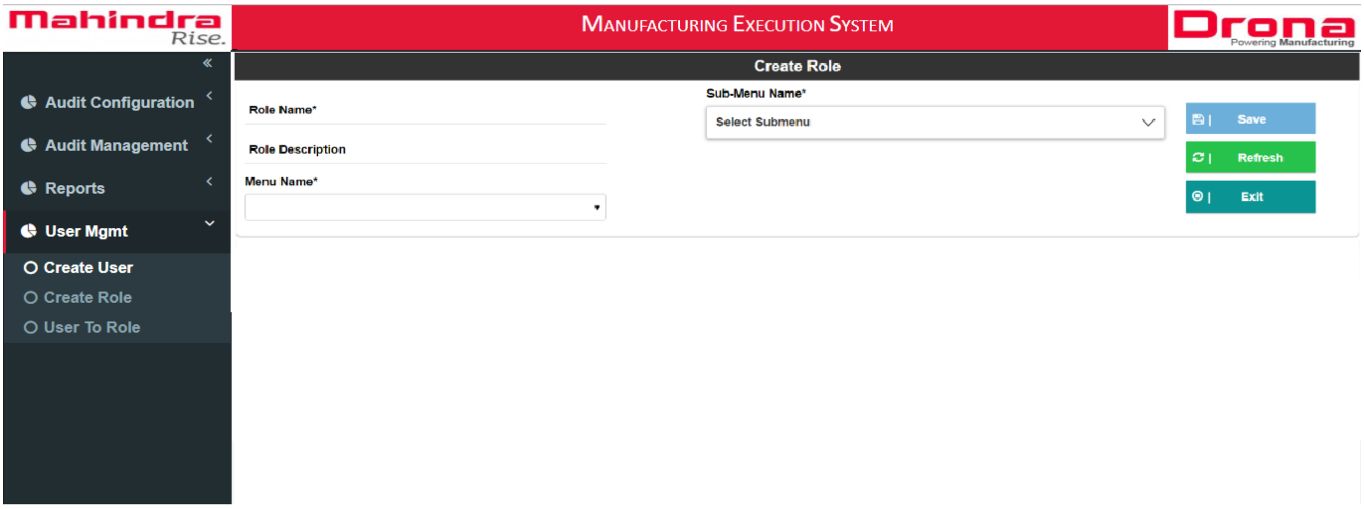
Description: User will enter the require fields.

- **User Name:** User will enter Auditor Name in text box.
- **Date of Birth:** User will Enter Date of Birth in text box.
- **Gender:** Against Auditor name, user will select gender from drop down.
- **Contact Number:** Against Auditor Name, user will enter mobile number.
- **Category (Department):** User will assign the category/department to user.

After entering required fields, user will click on the save button and data will be save in database.

4.17.2. Add User Configuration

This configuration is used to assign menu/sub menu to roles



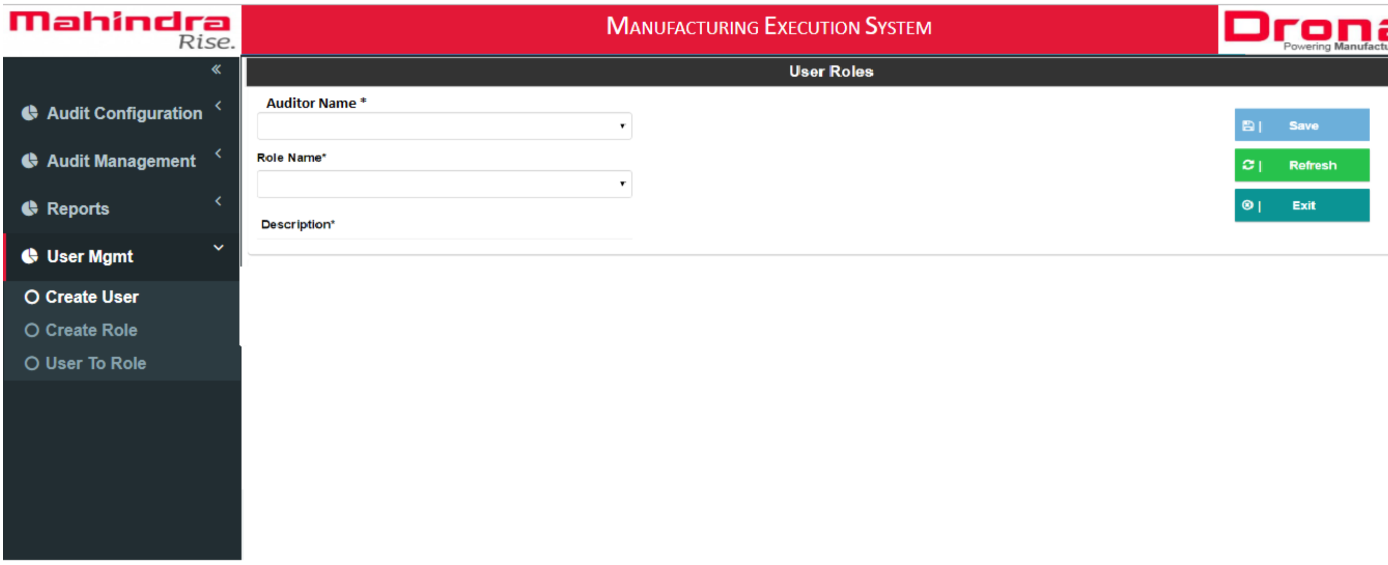
Description: User will enter the require fields.

- **Role Name:** User will enter Role name in text box.
- **Sub Menu Name**: Against Role name, User Will select Sub Menu Names.
- **Menu Name:** User will select Menu Name under Role Name.

After entering required fields, user will click on the save button and data will be save in data base.

4.17.3 User Role Configuration

This configuration is used to assign the user to the role.



Description: User will enter the require fields.

- **User Name:** User will enter Auditor name in text box.
- **Role Name:** Against Auditor name, user will enter Auditor ID in text box.
- **Description:** Against Auditor name, user will select gender from drop down.

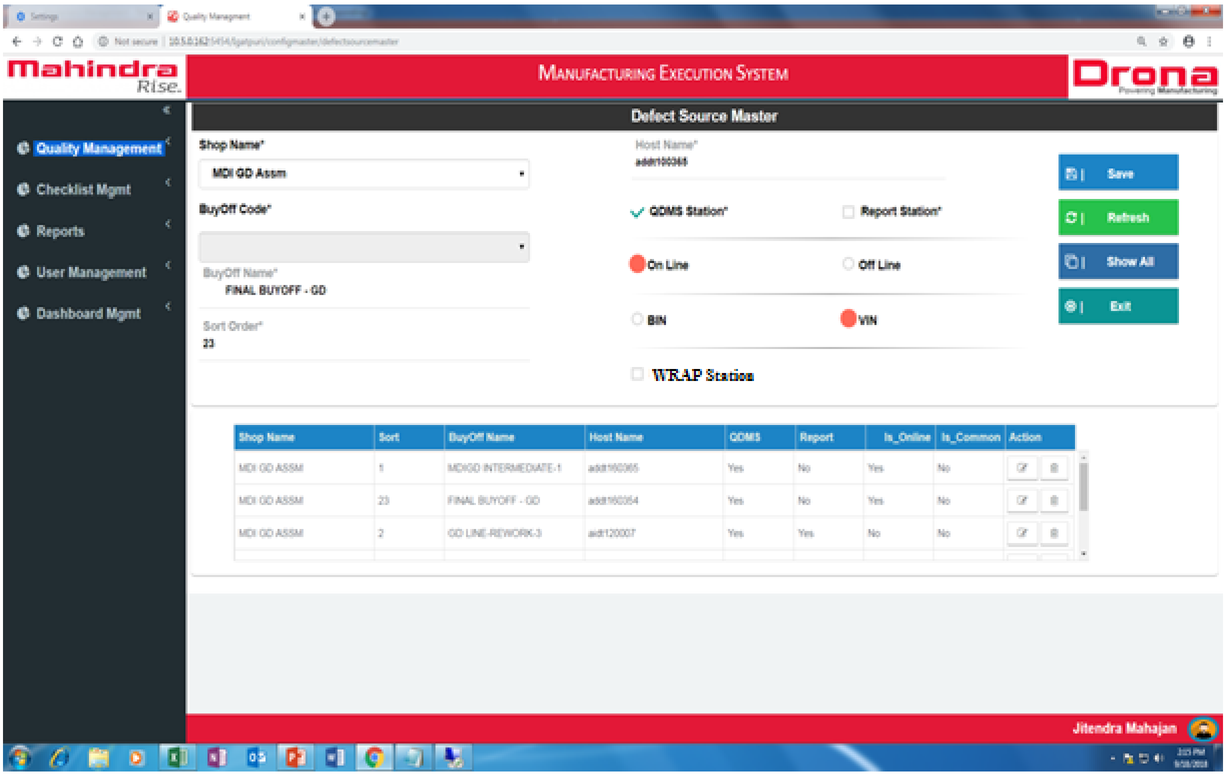
After entering required fields, user will click on the save button and data will be save in data base.

4.18 Master Data Configuration

Master data configuration is an important section to save the predefined data. This predefined data is used at the time of execution an application.

4.18.1 Defect Source Master

This functionality is used to define the QDMS station. Based on this configuration, QDMS client will open on production line. Below forms is used for Quality Defect Source Master Configuration.



Step 1: User will open Quality Defect Source Master Form.

Step 2: User will enter the required fields.

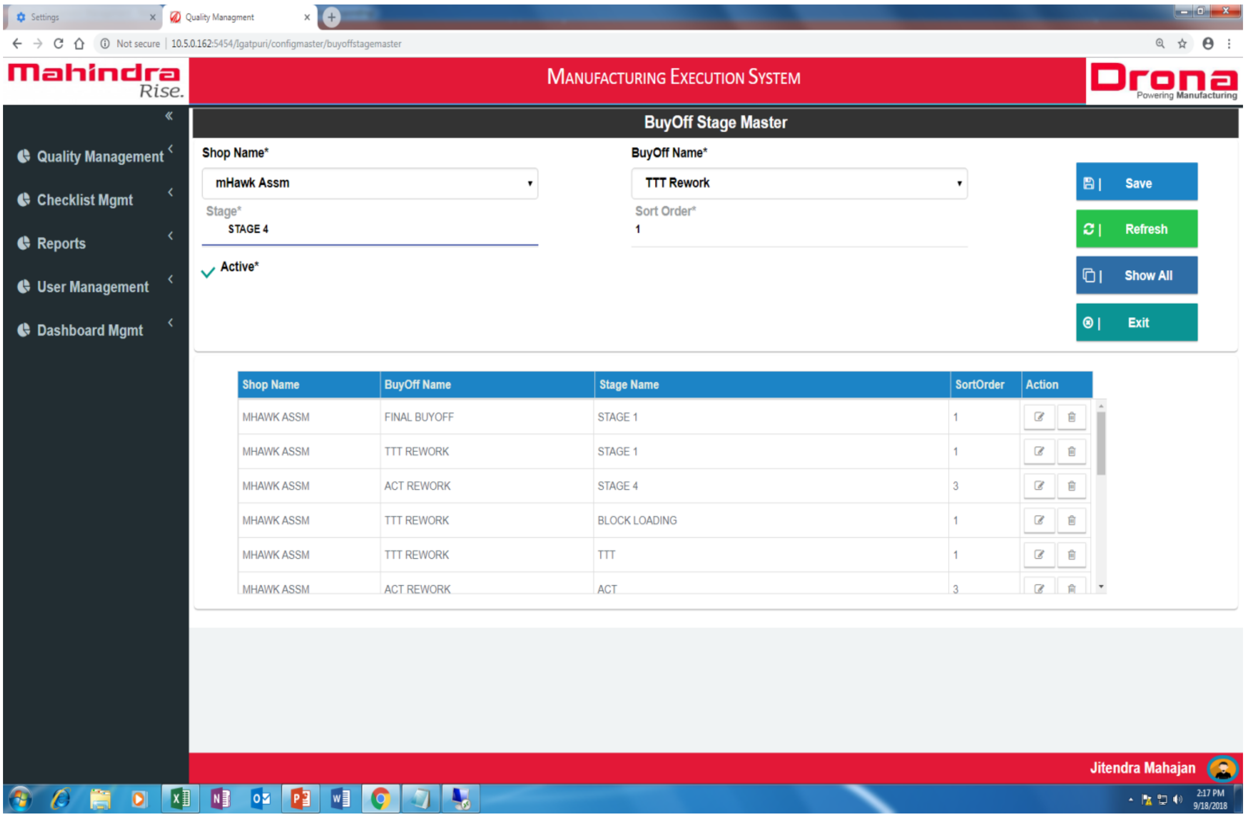
- Platform Name: User gets the lists of Platform added in platform configuration.
- Shop Name: User gets the lists of shop added in shop configuration.
- Buy off Code: It is an auto-generate field. After selection of line, this number gets generated. If configuration is not saved, then auto generated number will also not be saved in MES. This is used to ensure that there is no gap in buy off code.
- Sort Order: It is the sequence of QDMS client used on production line. Sort order for first quality client will be 1; for second client, it will be 2 etc.
- QDMS Station Radio Button: If the station configured is to be used as QDMS station, then the check box for QDMS is to be selected by user.
- Report Station Radio Button: User will select report station, against line, if required.
- Online Radio Button: If QDMS station is on the production line, then this option needs to be selected by user. Accordingly, QDMS client will appear on production station.
- Tester Line: If QDMS station is on the testing area, then this option needs to be selected by user. Accordingly, QDMS client will appear on production station.
- WRAP Station: If QDMS station is on the WRAP line, then this option needs to be selected by user. Accordingly, QDMS client will appear on WRAP station.

Step 3: After entering required fields then user will click on the save button then all data will be saved in database.

4.18.2 Quality Buy-Off Stage Master

This form is used to map stage against buy off station and line.

Step 1: User will open the Quality Buyoff Stage Master



Step 2: User will enter the require fields.

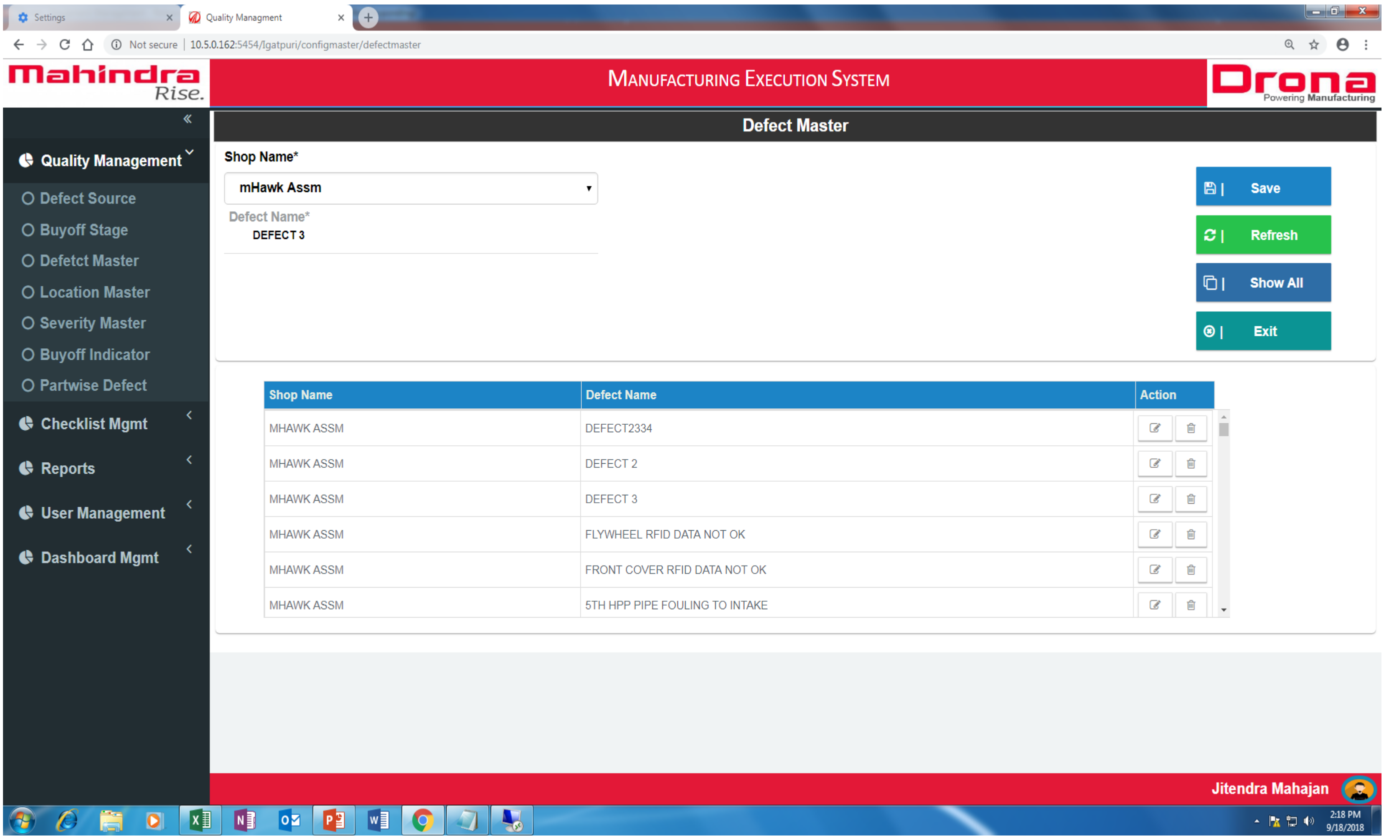
- Platform Name: User gets the lists of Platform added in platform configuration.
- Shop Name: User will select the Shop name from drop down.
- Buyoff: Against line, user will select buyoff from drop down. All QDMS defined stations will appear in this list.
- Stage Name: Against buy-off station, user will enter stage in textbox. All these stages will be mapped to the buy off station selected.
- Sort Order: This defines the sequence in which the defect related to entered stage will appear on QDMS screen.
- Active Check Box Button: Used to activate/de-activate the mapping done for the entered stage.

Step 3: After entering these fields, user will click on the Save button then all data will be saved in database.

4.18.3 Quality Defect Master

This functionality used to define the defects against line. Below forms is used for Quality Defect Master Configuration

Step 1: User will open Quality Defect Master tab



Step 2: User will enter the required fields.

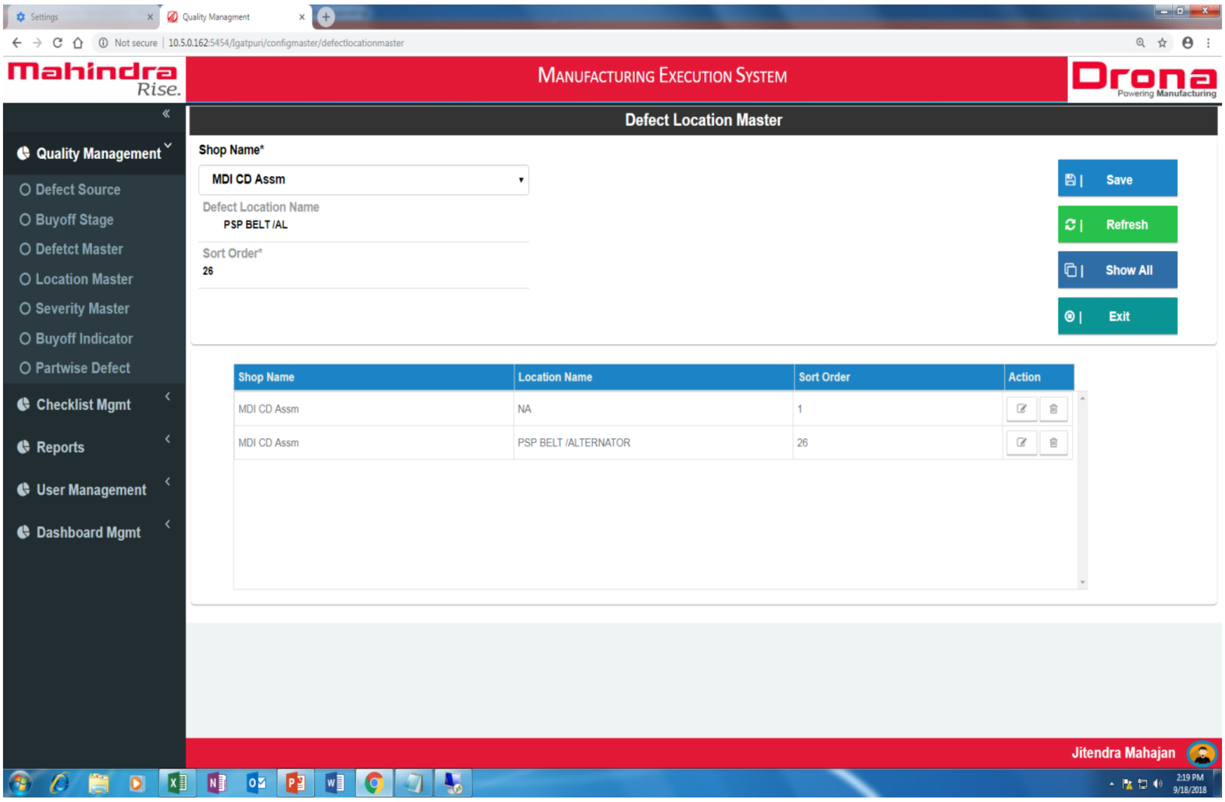
- Platform Name: User gets the lists of Platform added in platform configuration.
- Shop: User will select shop from drop down.
- Search Defect: Against line, user will enter defect in text box.

Step 3: After entering these fields, user will click on the add button and click on the save button then data will be saved in data base.

4.18.4 Quality Defect Location Master

This functionality is used to configure the location of defects. Below forms is used for Quality Defect Location Master Configuration

Step 1: User will open Quality Defect Location Master.



Step 2: User will enter the required fields.

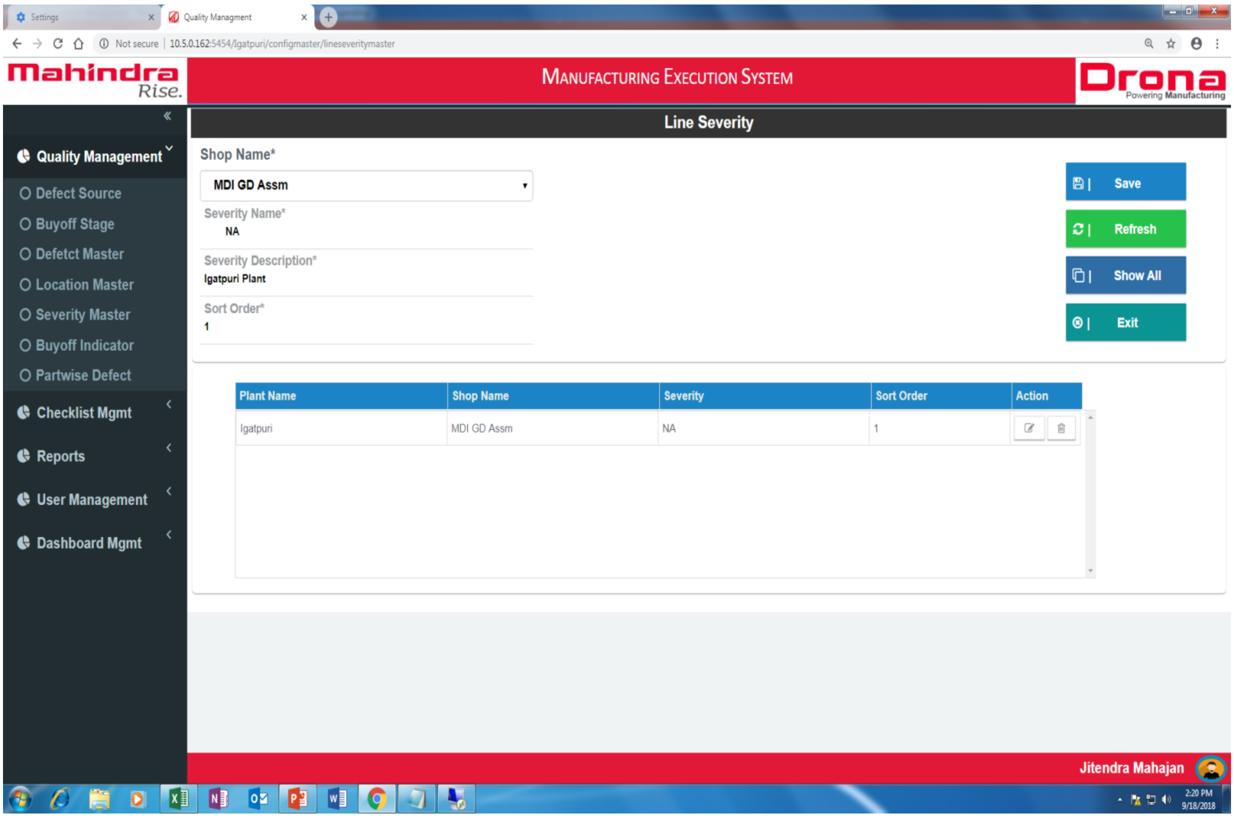
- Platform Name: User gets the lists of Platform added in platform configuration.
- Shop: User will select shop from drop down.
- Defect Location Name: Against to Line, user will enter defect location name in textbox LH or RH.
- Sort Order: Against defect location, user will enter sort order in textbox.

Step 3: After entering these fields, User will click on the save button and then data will be saved in database.

4.18.5 Quality Line Severity

This functionality used to define different level of defect severity. Below forms is used for Quality Line Severity Configuration

Step 1: User will open the Quality Line Severity form



Step 2: User will enter these fields

- Platform Name: User gets the lists of Platform added in platform configuration.
- Shop: User will select shop from drop down for which severity needs to be defined.
- Severity: Against line, user will enter severity in textbox.
- Sort Order: This defines the sequence in which severity will be displayed on QDMS client.

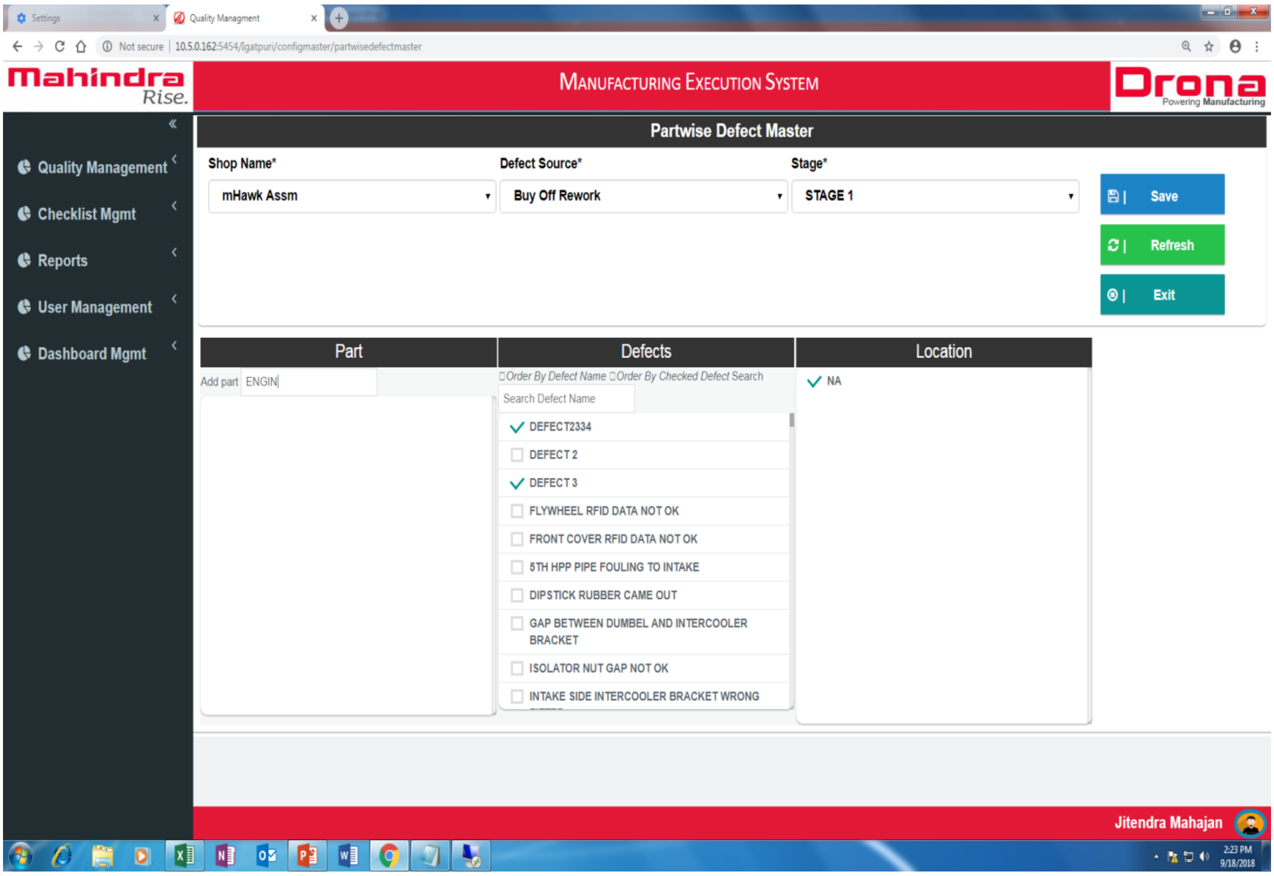
Step 3: After entering these fields the user will click on the save button and then data will be saved in data base.

4.18.6 Part Wise Defect Master

This functionality used to create the mapping of Shop, line, station/stage, defect source, part number and defects. Based on this mapping, defects appear on QDMS client against a part number.

Below forms is used for Part Wise defect master Configuration

Step 1: User click on the Quality Management module and in Quality management, user will click on the Part Wise defect master tab then Part Wise defect master index form will appear on QDMS screen.



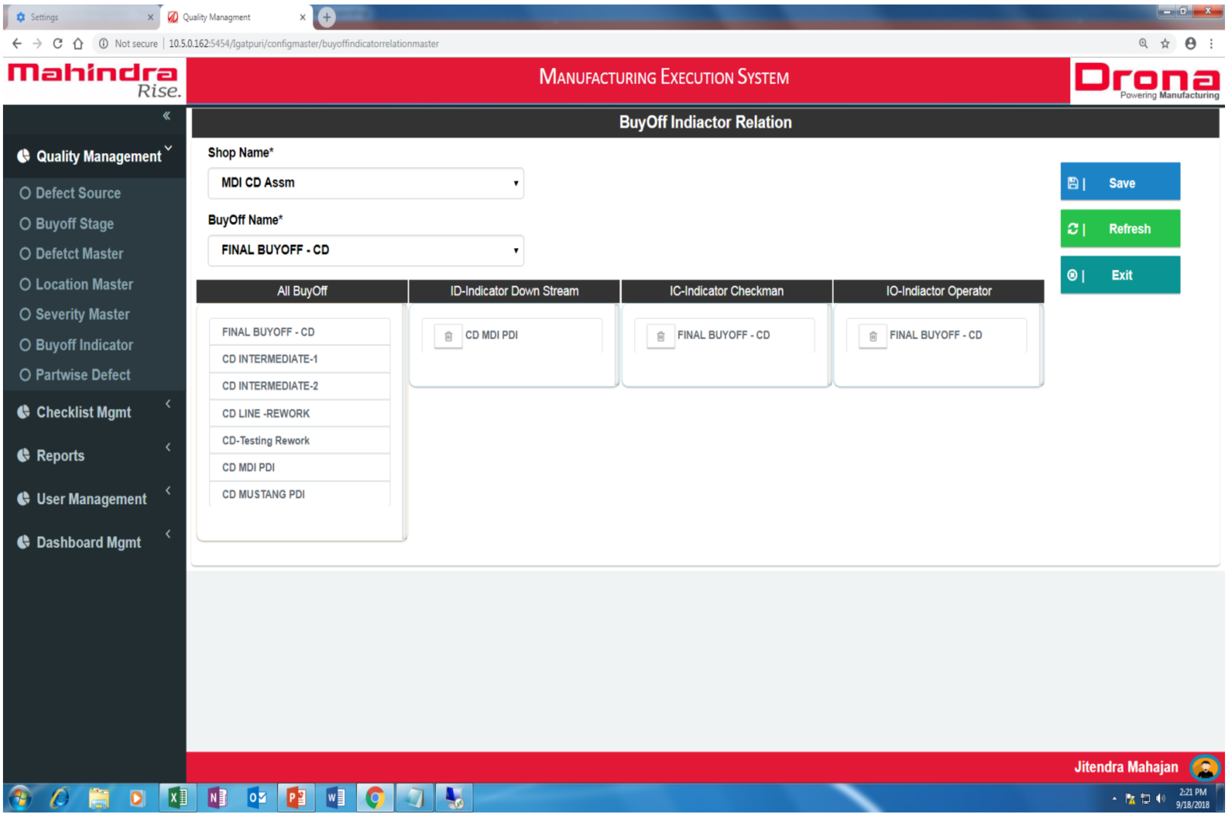
Step 2: User will enter these fields

- Platform Name: User gets the lists of Platform added in platform configuration.
- Shop: User will select shop from drop down.
- Defect Source: Against line, user will select defect source from drop down.
- Stage: Against line, user will select the stage from drop down.
- Add Part: Against line, defect source and stage, User will enter part in text box and against part, user will enter defects and location then click on the save button then data will be saved in database.

4.18.7 Buyoff indicator relation

Below forms is used for Buyoff Indicator Relation Configuration.

Step 1: User will open the Buyoff Indicator relation tab.



Step 2: User will enter required fields

- Platform Name: User gets the lists of Platform added in platform configuration.
- Shop: User will select the shop from drop down.
- Buyoff Code: Against to line, User will select buyoff code from drop down.
- All Buy Off: Against to line and buyoff code, all buy off station will be appeared in All buyoff grid.

There are three quality gates.

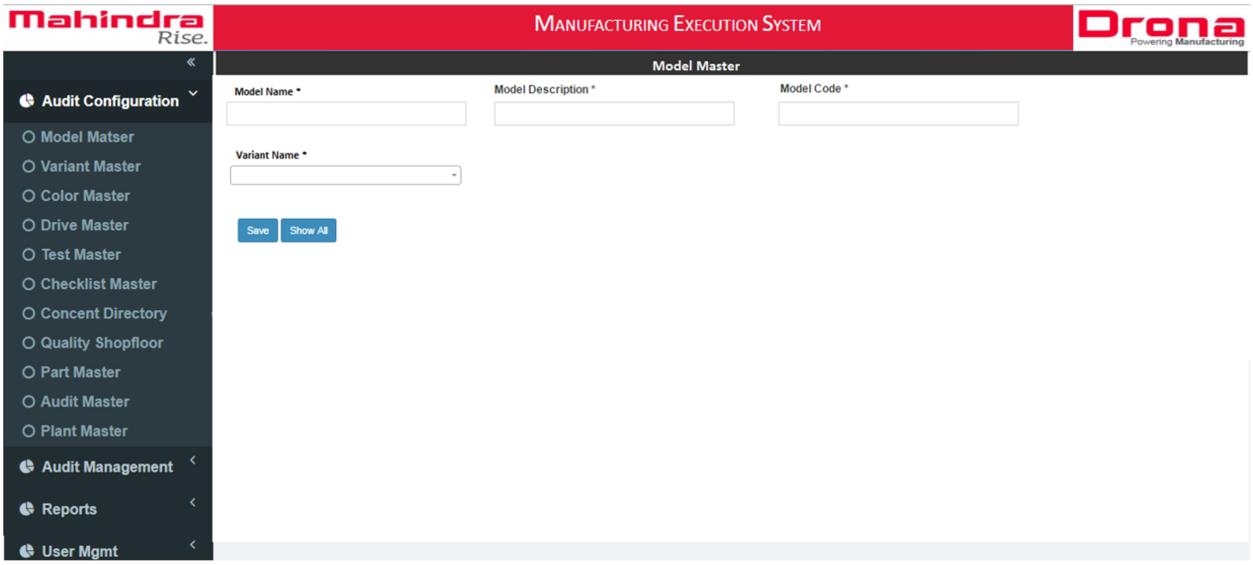
1. ID - indicator down stream
2. IO - indicator operator
3. IC - Indicator check

Step 3: User will click on the Quality gate radio button that buy off station will be appeared in Quality gate and user will click on the save button after save button data will be saved in data base.

4.18.8 Model Master Configuration

This form used to store the model wise master data

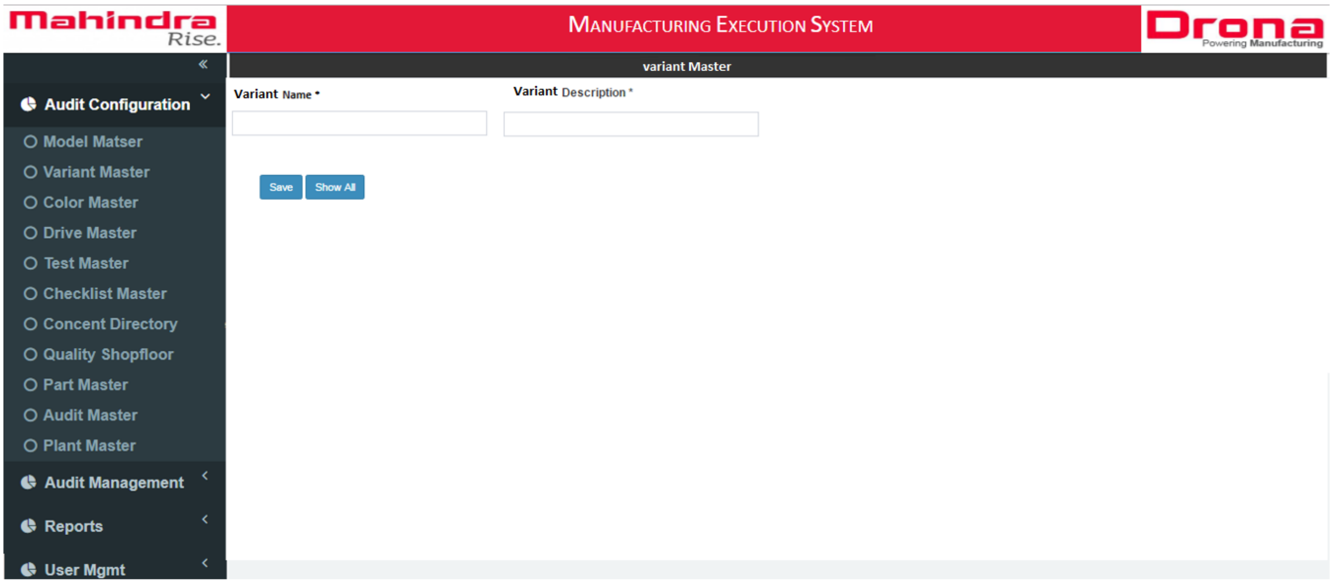
Description: - User should enter all required fields. After clicking on save buttons, the master data will get stored in the database.



4.18.9 Variant Master

This form used to store vehicle Variant list e.g. Diesel, Petrol, CNG etc.

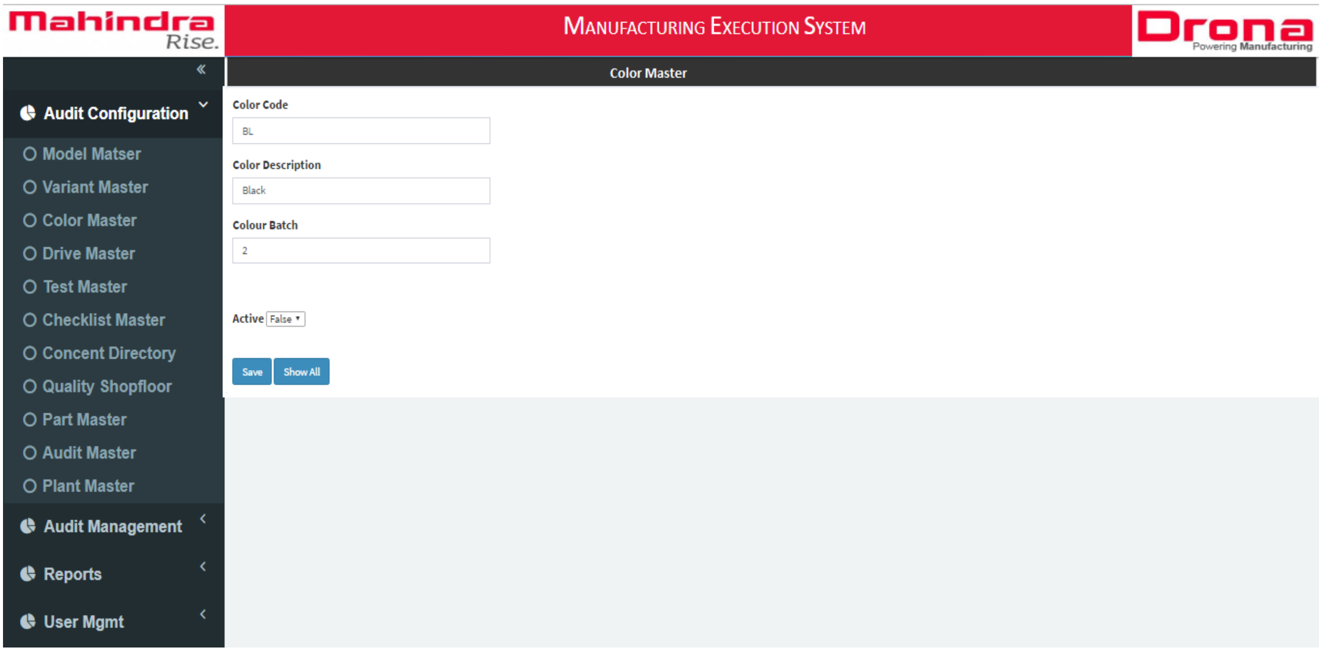
Description: - User should enter all required fields. After clicking on save buttons, the master data will get stored in the database.



4.18.10 Color Master

This form used to store different colors master data

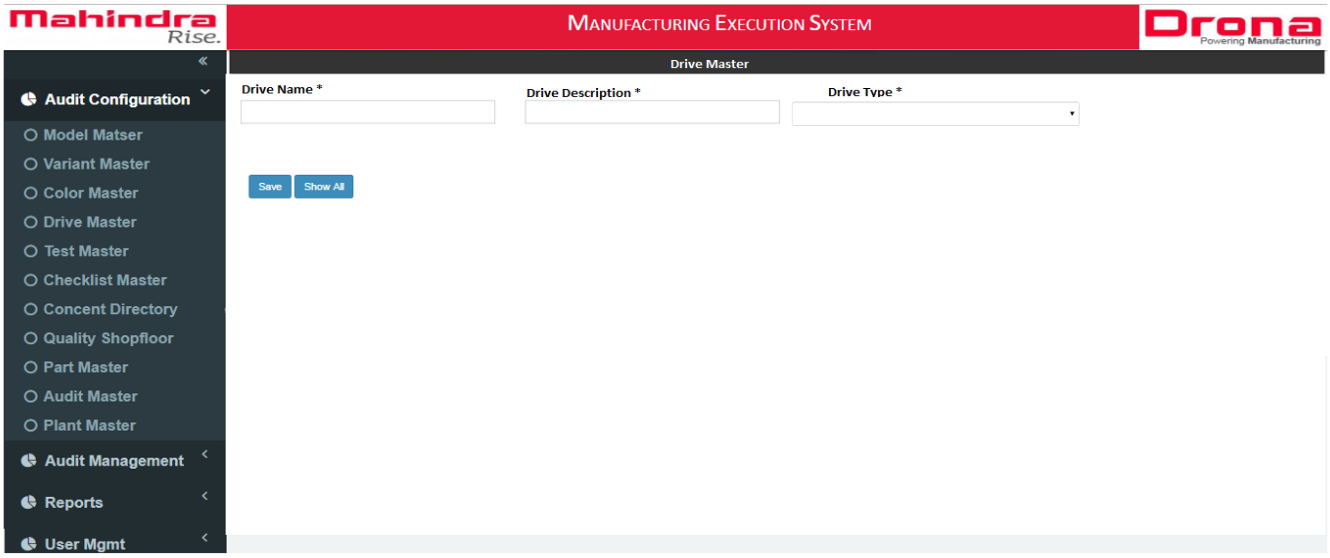
Description: - User should enter all required fields. After clicking on save buttons, the master data will get stored in the database.



4.18.11 Drive Master

This form used to create drive master like LHD, RHD

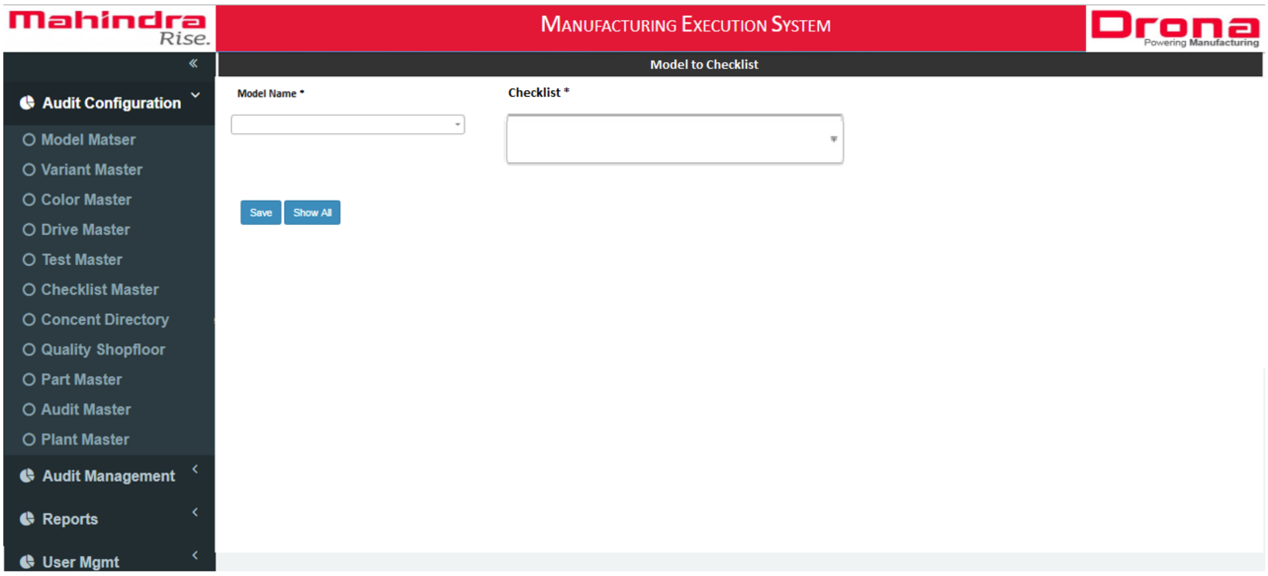
Description: - User should enter all required fields. After clicking on save buttons, the master data will get stored in the database.



4.18.12 Model to Checklist Master

This form used to store checklist against model.

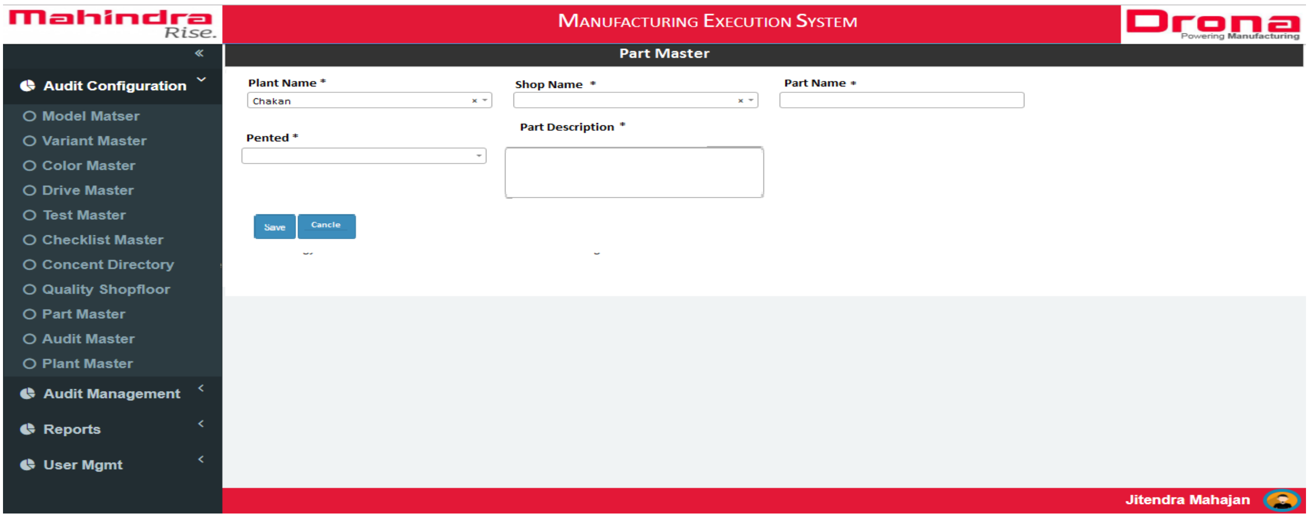
Description: - User should enter all required fields. After clicking on save buttons, the master data will get stored in the database.



4.18.13 Part Master

This form is used to store parts related data

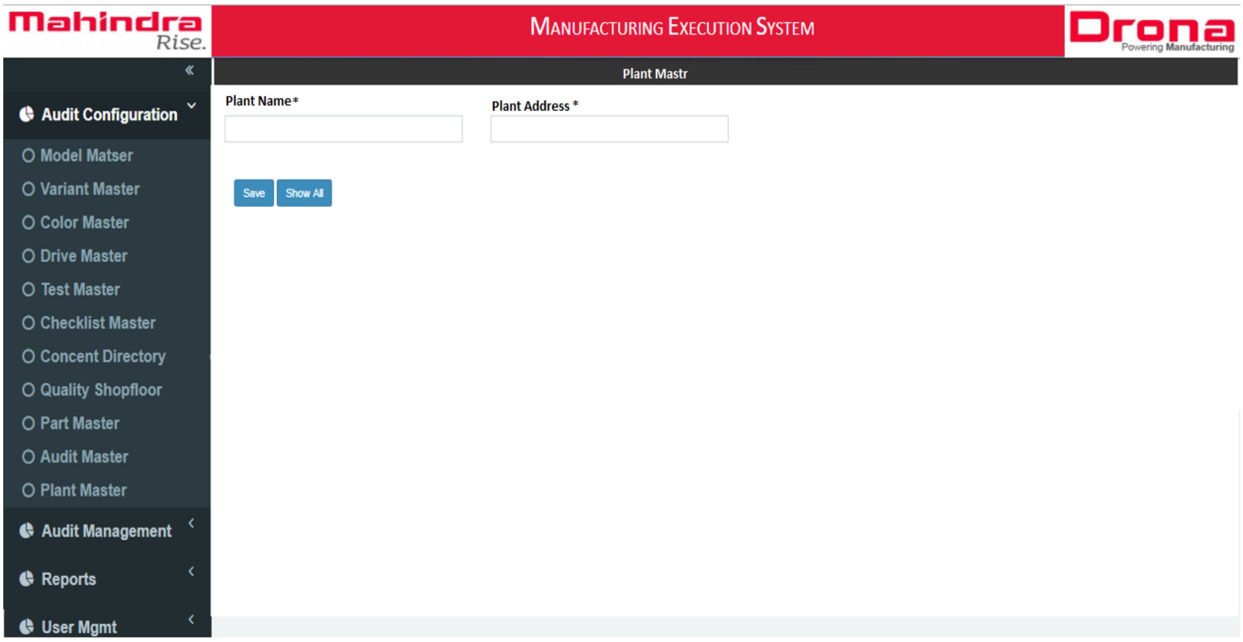
Description: - User should enter all required fields. After clicking on save buttons, the master data will get stored in the database.



4.18.14 Plant Master

This form used to store Plant related data

Description: - User should enter all required fields. After clicking on save buttons, the master data will get stored in the database.

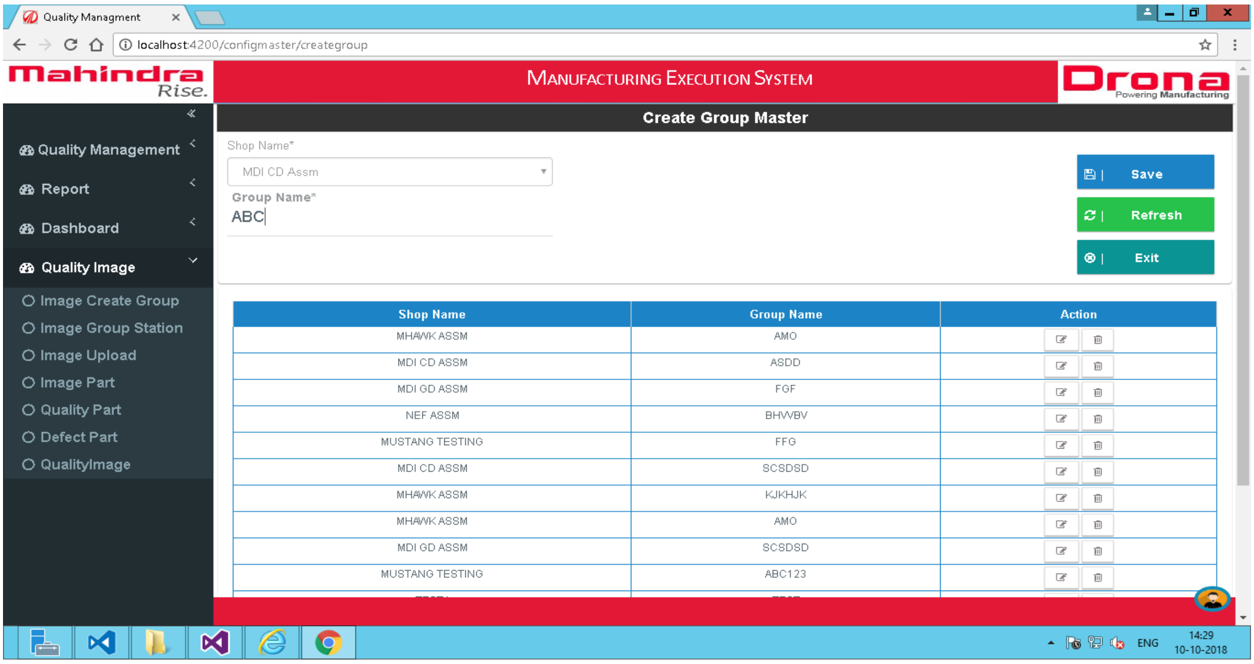


- **Plant Name:** User Enter the plant name of under this project that all name saves in drop down of plant Name.
- **Plant address:** user will enter the details of plant address.

4.18.15 Image Group Master

This form used to Configuration allow to define group of images for shop which consist of number of images by views

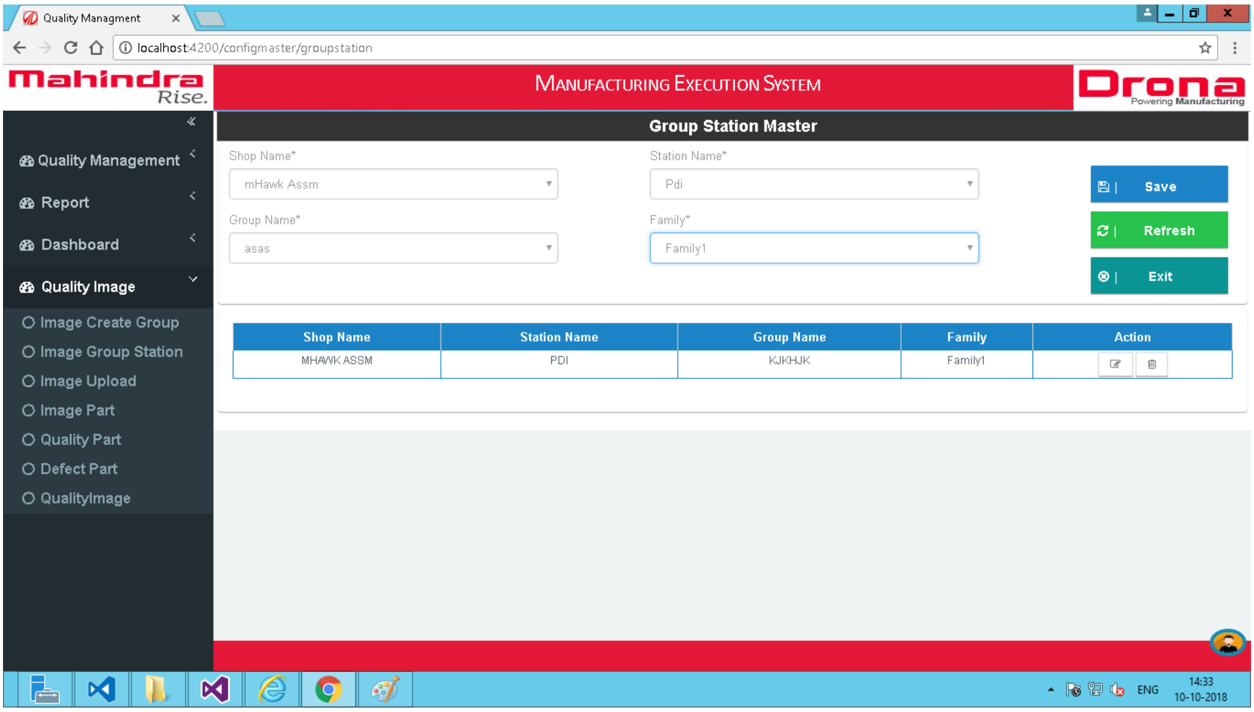
Description: - User should enter all required fields. After clicking on save buttons, the master data will get stored in the database.



- **Group Name:** Define name of group
- **Platform Name:** User gets the lists of Platform added in platform configuration.
- **Shop ID:** Select shop for defining image group
- **Plant ID:** Select plant for defining image group

4.18.16 Quality Image Group to Station

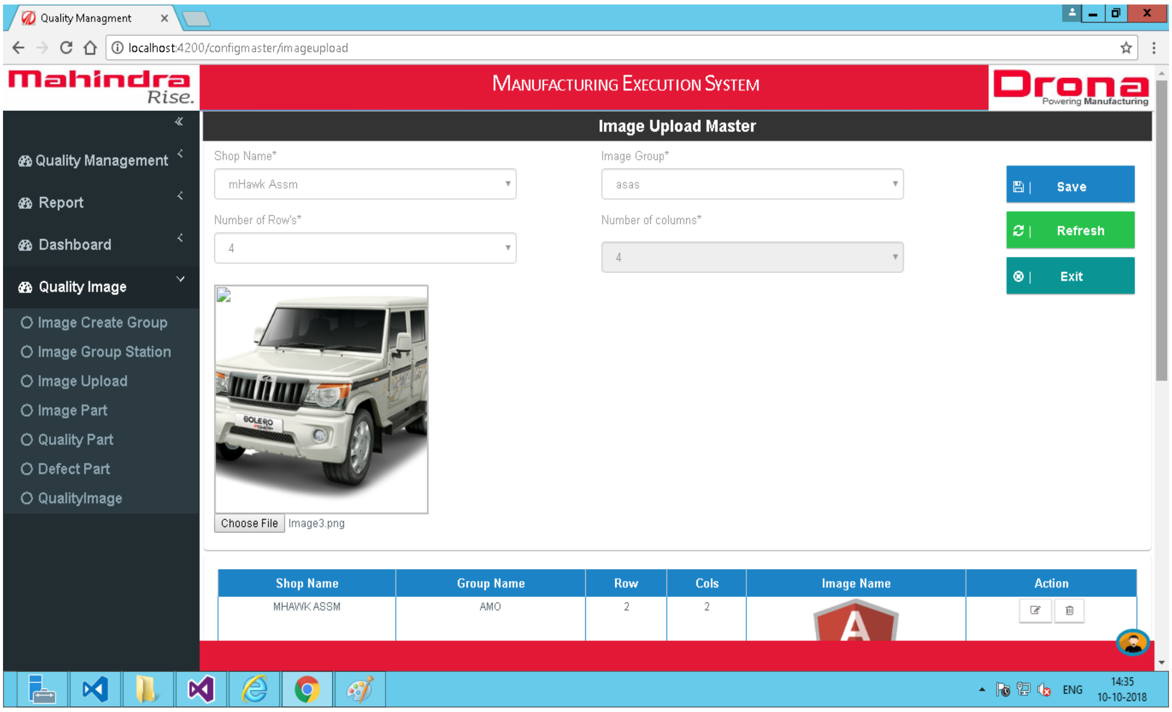
Configuration is used to assign the image group to station by family. The allocation is required which will display images on QP screen when operator scan configured family vehicle on station.



- **Platform Name:** User gets the lists of Platform added in platform configuration**.**
- **Shop Name:** Select Shop Name to get line
- **Line Name:** Select Line Name to get station
- **Station Name:** Select Station to assign image group
- **Group Name:** Select Group Name which will be assigned to station
- **Family:** Select family for which selected group images will appear

4.18.17 Quality Images

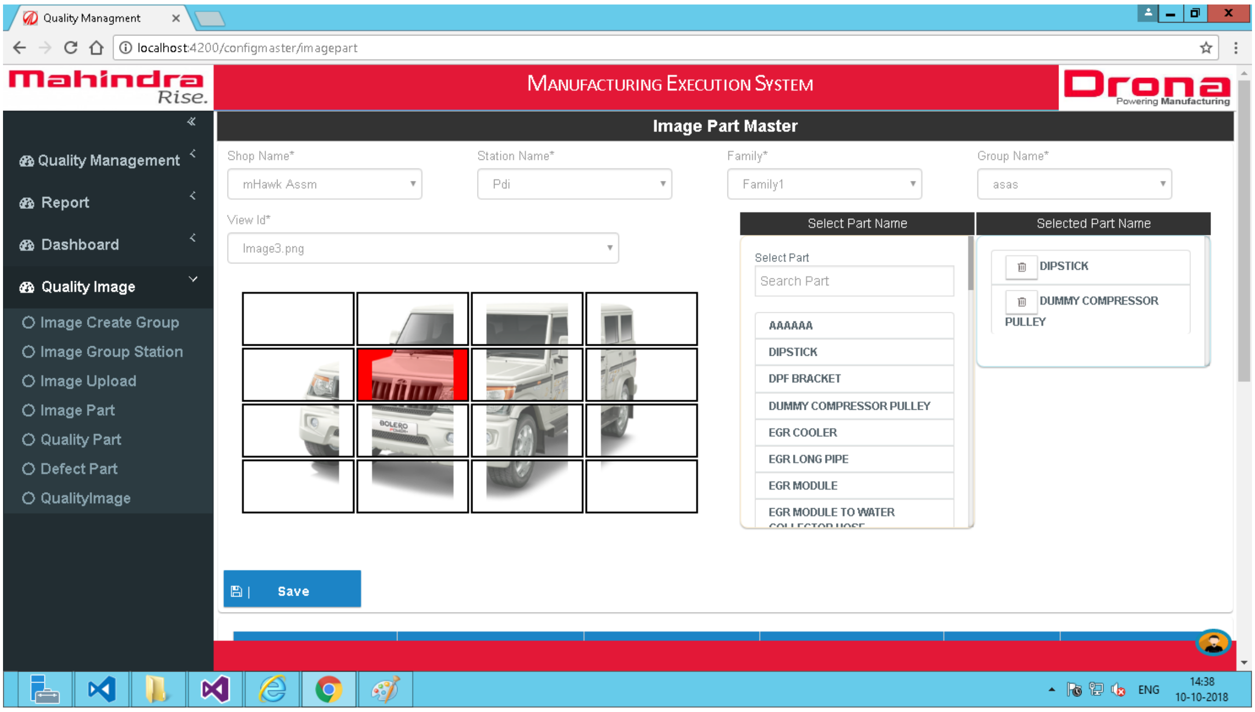
Configuration allow user to upload the images for different views of groups



- **Platform Name:** User gets the lists of Platform added in platform configuration.
- **Shop Name:** Select Shop Name to get image group
- **Choose file:** Select quality image to upload from local PC
- **Upload:** Upload selected image in QDMS

4.18.18 Quality Parts to Image

Configuration is used to assign parts to quality images



- **Platform Name:** User gets the lists of Platform added in platform configuration**.**
- **Shop Name:** Select Shop to get line and family
- **Line Name:** Select Line to get station
- **Station Name**: Select station to get group name
- **Group Name:** Select group to get view
- **View ID:** Select view to get view image
- **Select part:** List of un-assigned part to image
- **Selected part name:** List of assigned part to image

4.19 Report Management Configuration:

There are multiple reports need to be provided to business users. Following are the reports need to be implemented

BIW

1. FRC /FBO – ShiftWise, day wise, Month wise, year wise, BWT Wise
2. Top 5 concerns – Occurrence wise + Severity*Occurrence wise – daily ShiftWise
3. TCF & Paint Shop RPT related to BIW – Daily, Shift wise, Monthly, Yearly
4. RFI DPT
5. Concentration diagram
6. Attribution wise concern status
7. Rework Manhours
8. Report for number of hold vehicles in the respective areas

Paint Shop

1. Top 5 concerns – Occurrence wise + Severity*Occurrence wise – daily ShiftWise
2. TCF RPT related to Paint Shop – Daily, Shift wise, Monthly, Yearly
3. RFI DPT
4. Concentration diagram
5. Attribution wise concern status
6. Rework Manhours
7. Report for number of hold vehicles in the respective areas
8. Hold Bodies status model wise, date wise, ShiftWise monitoring

TCF

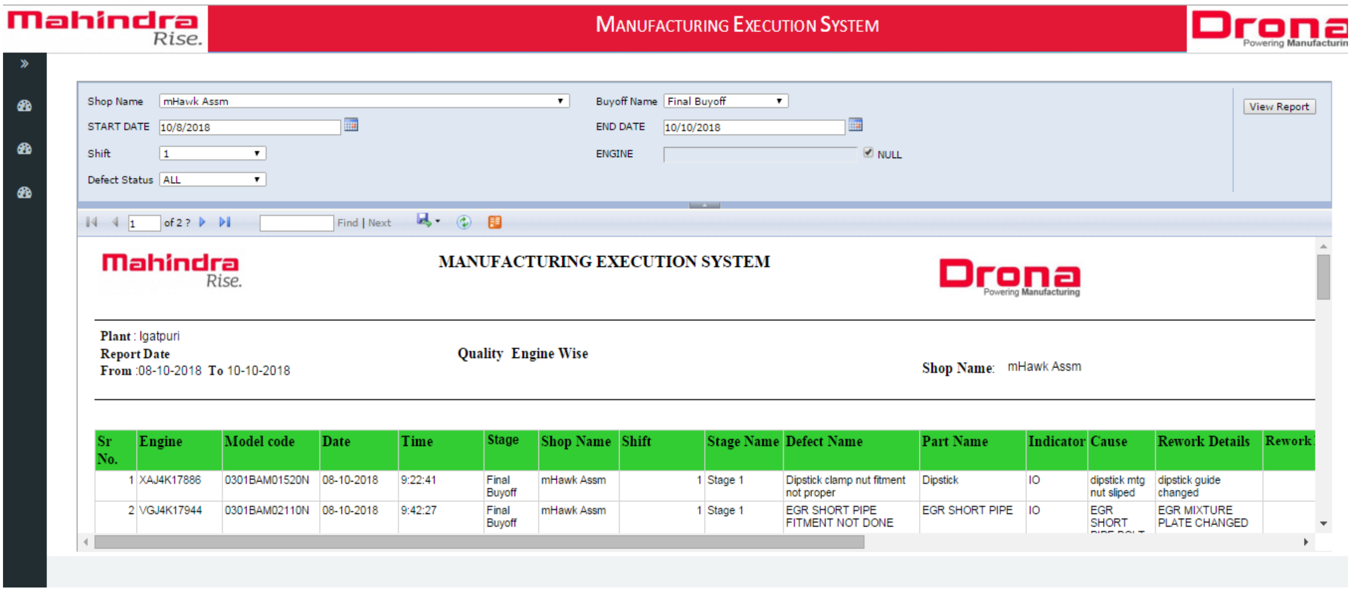
1. FRC /FBO – ShiftWise, day wise, Month wise, year wise, BWT Wise, Line wise, model wise
2. BIQ Pillar wise report
3. Top 5 concerns – Occurrence wise + Severity*Occurrence wise – daily ShiftWise
4. RFI DPT
5. Total RPT (up to WRAP)
6. Offline RPT
7. Offline RPT (Functional/Nonfunctional)
8. Shower FRC
9. Concentration diagram (Shower + Paint booth)
10. Attribution wise concern status
11. Rework Manhours
12. Report for number of hold vehicles in the respective areas

RFI

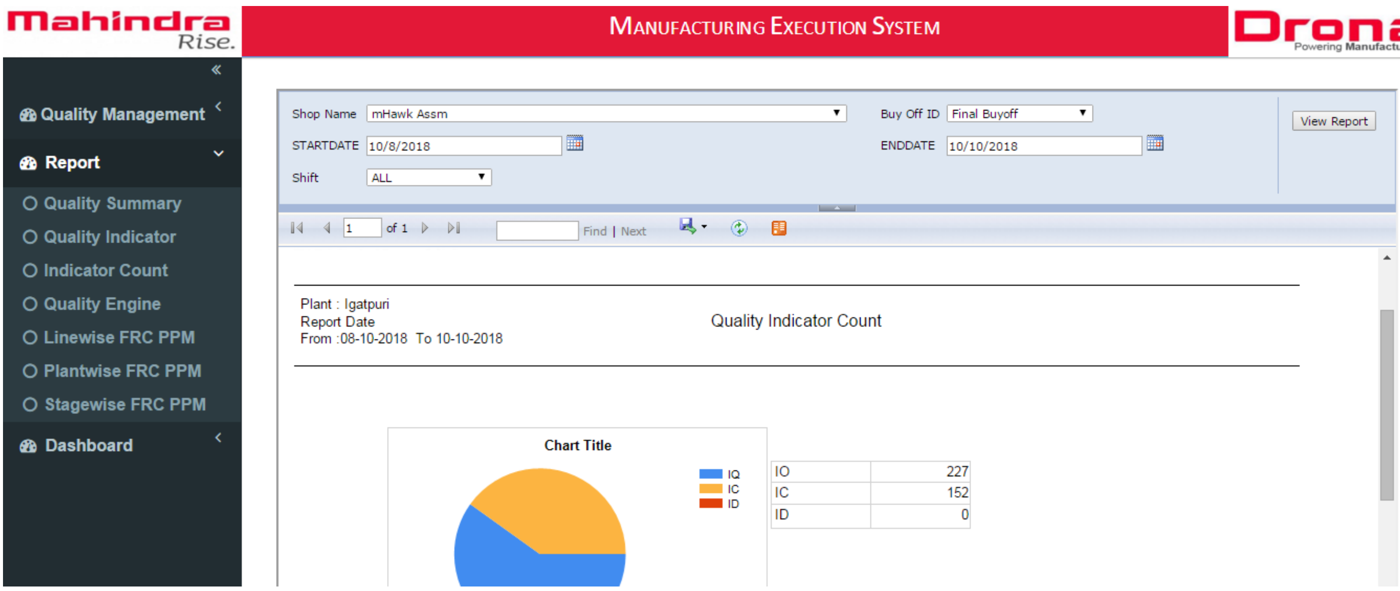
1. Weighted DPV as per model, Within model- department wise, within attribution wise and model wise
2. Same above only for V1&V1+

Following are some sample screens which have been implemented currently for Igatpuri QDMS. We will implement the reports for this project in a same way.

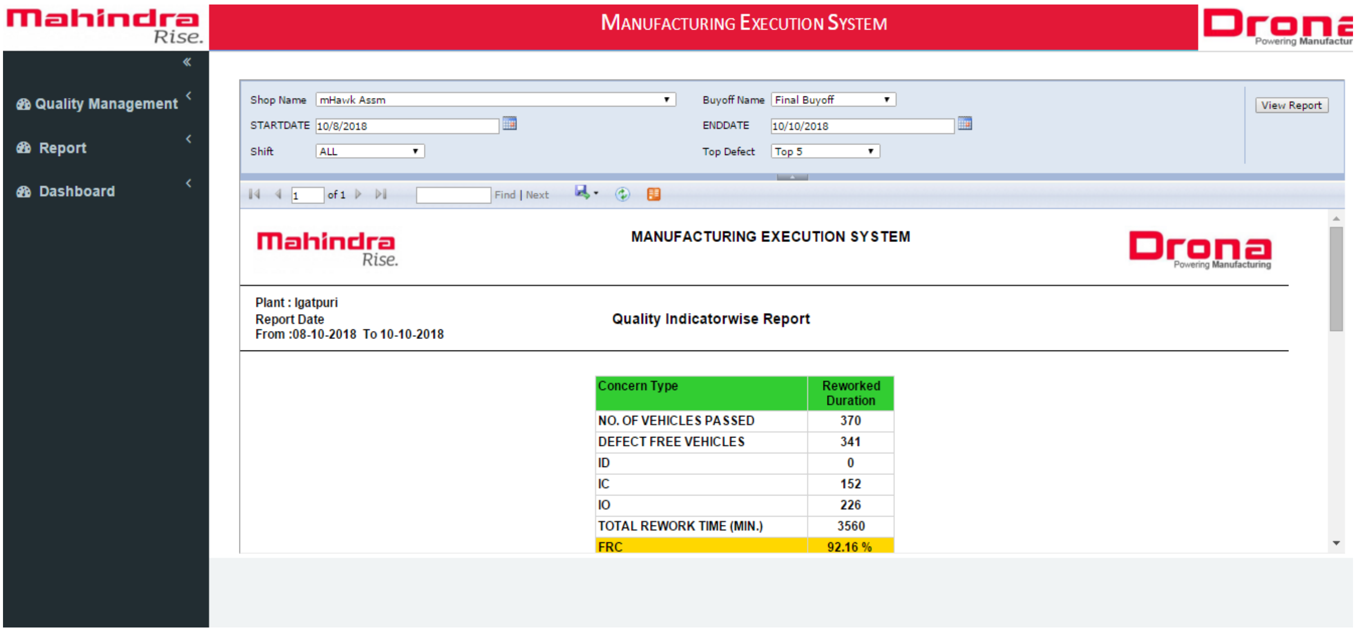
Engine Wise Defect Report



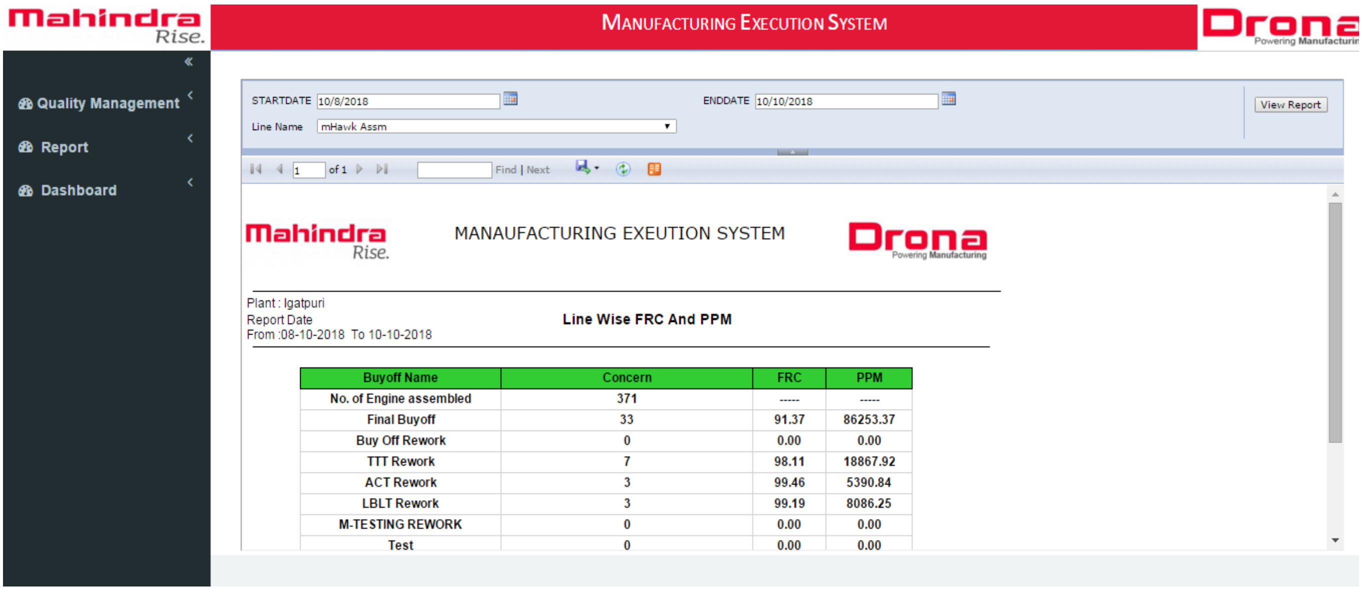
Quality Indicator Count



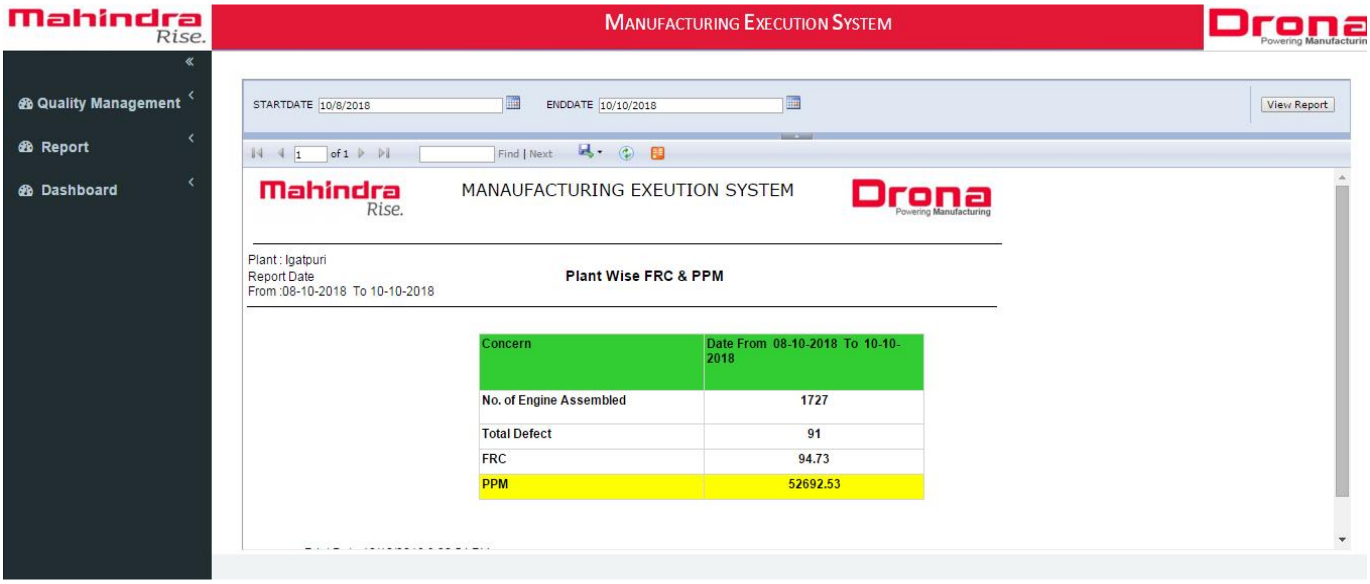
Quality Indicator wise Report



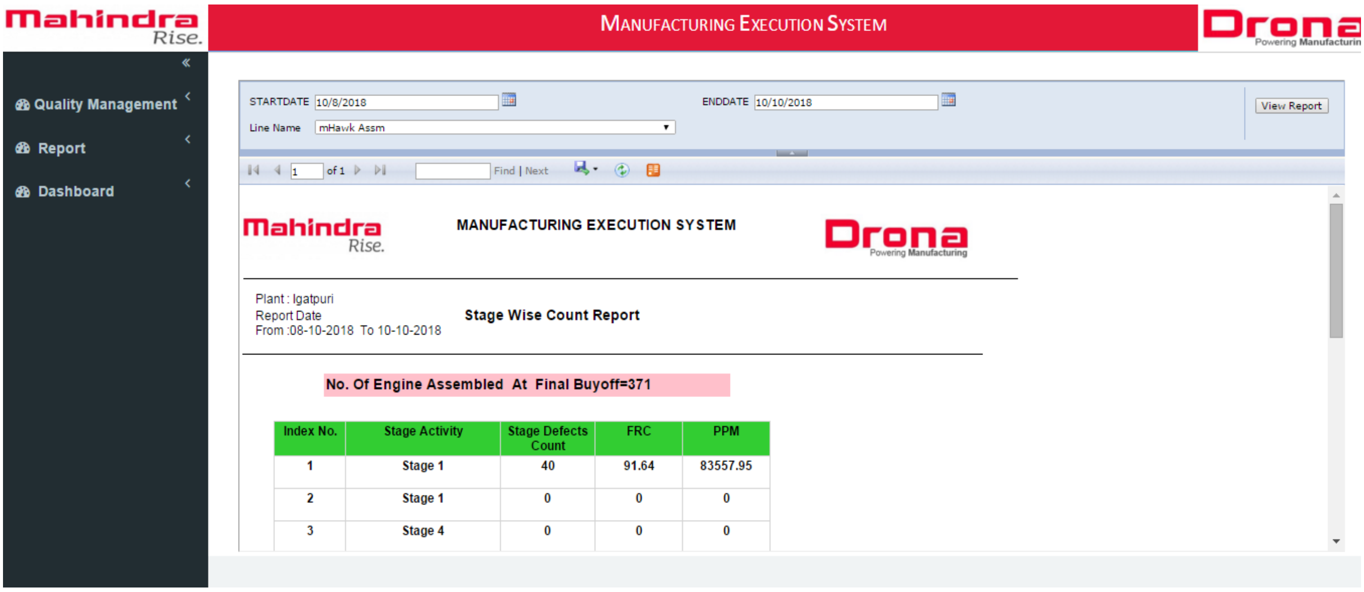
Line wise FRC and PPM Report



Plant wise FRC and PPM Report



Stage wise FRC and PPM Report



Quality Summery Report

Date	Stage	Shift	Total Production	No Defect	Reworked	FRC	FBO
08-10-2018	Final Buyoff	1	7	0	7	0.00	100.00
08-10-2018	Final Buyoff	2	0	0	0	0.00	0.00
08-10-2018	Final Buyoff	3	0	0	0	0.00	0.00
08-10-2018	Buy Off Rework	1	0	0	0	0.00	0.00

5. Sign-Off

Approval Date:

Approval Organization	Approved by (Name)	Designation	Signature